Supplementary information: Scalable Dataset Splitting through Constrained Low-Rank Similarity Optimization

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1 Current-version evidence and provenance

The primary article describes v0.2.0.dev3, with numerical package source pinned to commit `f7e0866`. The 199-case primary grid yields 191 successful numerical cases and eight native DataSAIL configurations reporting infeasibility. The v1 benchmark adapter subsequently raised an exception when reading those absent native assignments; their solver evidence is preserved and no score is imputed. A separate 72-case publication check uses structured infeasibility outcomes, equal-confidence accuracy controls and native-memory profiles. Full-kernel fidelity supplies 900 comparisons over six datasets, three seeds, two powers, five landmark counts and five partitions.

Every successful case retains its source/input manifest and assignment checksums. Independent full NumPy Tanimoto calculations check 246 saved global and local scores, including all group controls and current external-method comparisons. The package suite passes 223 tests. These checks establish implementation and artifact consistency; they do not certify an optimizer or a deployment boundary.

FreeSolv, ESOL, BACE, BBBP, Lipophilicity and Tox21 are full public datasets with all successfully featurized entities retained. QM8 additionally tests audit budgets. These datasets have informed development and are not described as permanent holdouts. Optimization uses feature values and declared constraints, never model test outcomes. Exact input identities and feature counts are recorded in the manifests.

Original thresholded and alternative-fingerprint outputs remain in the per-case records. Raw mean nearest similarity was selected for the current presentation during revision to keep an interpretable common scale. This presentation choice did not change the fixed method settings or model recipes. The analyses are descriptive development studies, not preregistered confirmatory evaluations.

2 Factor construction, optimizer and feasibility contracts

The landmark matrix is diagonalized in float64. Eigenvalues at or below a relative threshold of the largest eigenvalue are discarded rather than floored; the default relative tolerance is 10^{-8} with a numerical-size safeguard. The effective factor dimension can therefore be smaller than the requested landmark count. Entity–landmark products are row blocked.

The similarity-spreading landmark heuristic, historically named `kmeans++`, samples distinct eligible rows with weights proportional to $(1 - \max_j S_{ij})^2$, falling back to uniform unselected rows when all residual weights vanish. It is not ordinary squared Hilbert-distance sampling. Ordinary k-means++ guarantees are not claimed for this rule. Uniform landmark sampling remains available. Established work discusses alternative sampling strategies [Kumar et al., 2012, Musco and Musco, 2017].

For nonnegative requested fractions q_c summing to one, largest-remainder rounding gives targets summing to n. The tolerance corridor is formed from $(1 \pm \epsilon)nq_c$, with integer bounds widened only enough to contain the rounded target. For point-level assignments, the two-fold linear proposal sorts preference differences within both folds' bounds. More than two folds use a regret-ordered capacity peel. An actual factor-objective check rejects worsening proposals. The best unrefined candidate is retained while two candidates are refined before final selection. Exact capacities use count-preserving swaps; corridors permit legal single moves. The final objective guard can reject numerically worsening refinement. Point refinement is skipped above two million entities. These are heuristic choices, not convergence or global optimality certificates.

The standard combined pipeline also offers the pointwise solution as a grouped warm start. The controlled same-group study instead supplies the same seeded feasible group packing to both power settings. Packing preferences are independent of factor values. Exact-capacity feasibility can use a bounded integer-program fallback. An independently checked integer incumbent is accepted; a proved infeasible result and a timeout without a valid incumbent are reported separately. The strict policy never dissolves a group. The count-first policy may dissolve groups to meet counts, and records those relaxations.

Optional fold-pair weights enter factor optimization and weighted scoring consistently. All primary comparisons use equal weights on distinct fold pairs. The n-D software path uses sparse identity factors and record-weighted feature preparation; these features are tested software capabilities outside the primary molecular comparison. Their API and memory scopes are documented with the package.

3 Exact-reference decomposition and matched-group controls

The small reference suite contains twenty binary 12×24 matrices drawn with Bernoulli probability 0.3 at fixed seeds. Counts are 8/2/2, not the larger datasets' 80/10/10 proportions. Enumerating 2,970 unrestricted assignments or thirty assignments of six paired groups gives exact minima for both the full and approximate kernels. For each returned partition, the records retain heuristic suboptimality for the approximate kernel, the largest approximation error over the feasible set, and suboptimality for the full kernel. These finite enumerations do not certify larger problems.

Table 1: Exact small-instance references. Each value is the maximum over twenty 12-entity instances, divided by the true kernel's off-diagonal mass. Here δ_{enum} is the maximum cut-value error over the enumerated feasible set, η is heuristic suboptimality for the factor objective, and True gap is suboptimality for the original kernel. Counts are 8/2/2; paired groups have two entities each.

Groups	Power	Landmarks	δ_{enum}	η	True gap
None	1	2	0.5326	0.0000	0.1574
None	1	4	0.2771	0.0028	0.1093
None	1	8	0.0839	0.0207	0.0414
None	1	12	0.0000	0.0132	0.0132
None	3	2	0.7621	0.0000	0.3290
None	3	4	0.5588	0.0376	0.3656
None	3	8	0.2947	0.0334	0.2051
None	3	12	0.0000	0.0408	0.0408
Pairs	1	2	0.4591	0.0000	0.1042
Pairs	1	4	0.2721	0.0000	0.0767
Pairs	1	8	0.0839	0.0000	0.0593
Pairs	1	12	0.0000	0.0000	0.0000
Pairs	3	2	0.6702	0.0000	0.3286
Pairs	3	4	0.5532	0.0000	0.3178
Pairs	3	8	0.2947	0.0000	0.2609
Pairs	3	12	0.0000	0.0000	0.0000

The full-dataset group controls retain one identical group array and exact counts for the feasible control and both optimized variants. The group arrays and every assignment are directly checked for preservation. The main table averages three seeds within each of six datasets. These are matched mechanism controls; native DataSAIL, Lo-Hi or scaffold workflows can impose different admissible partitions or retention contracts.

4 Accuracy allocation and uncertainty

The cut audit samples a signed residual mean, takes its absolute value, and normalizes by mean absolute off-diagonal kernel mass. This is cut error for the specified partition, not entrywise mean absolute error or error relative to the cut itself. Bounded empirical-Bernstein intervals cover numerator and denominator components jointly [Maurer and Pontil, 2009]. The two-sided mean radius for samples in $[a, b]$ is

$$\sqrt{\frac{2\widehat{v}\log(4/\alpha)}{m}} + \frac{7(b-a)\log(4/\alpha)}{3(m-1)},$$

with unbiased sample variance $\widehat{v}$, sample count $m \geq 2$ and component failure budget α. It applies the cited one-sided bound to both tails after range scaling. Analytically zero contributions can be excluded; sampled zero mass alone never establishes an empty or accurate

factor. A denominator interval containing zero gives an unestablished result. Known-small populations are enumerated exactly.

For the optional joint rank/probe policy, a finite doubling grid defines at most RQ possible checks. Each selection check spends failure probability $0.025/(RQ)$, and a separately sampled final audit spends 0.025. The union bound does not require independence between selection checks. A candidate partition at a fixed rank is computed without using the audit pairs. When an interval straddles tolerance, probe count grows first; when error is established or probes reach their limit, rank grows if possible. The reached probe count carries forward. All pair evaluations, including repeated checks, are counted. The historical diagnostic key `ci95` contains the interval at its accompanying reported confidence, which can exceed 95 percent.

Fixed-probe controls use the same joint-valid policy with equal initial and maximum probe counts. This avoids comparing policies with different overall confidence budgets. Selection and final auditing must both establish tolerance; exhaustion remains visible even when the point estimate looks favorable. The older fixed-probe experiment, whose two stages separately used 95 percent coverage, remains in its original records but supplies no equal-confidence efficiency claim.

Table 2: Equal-confidence accuracy/cost controls. Every policy uses the same 95 percent joint selection/final-audit coverage, cut tolerance 0.05, rank range 32–512, two refined candidates and exact counts. A fixed policy is a single-size probe grid, not the older policy with separate unadjusted 95 percent checks. Established requires both selection and independent final audit to establish tolerance. These datasets are development evidence; no model outcome selects a policy.

Dataset	Probe policy	Median call (s)	Ranks	Established
FreeSolv	Fixed 2k	0.84	512–512	0/3
FreeSolv	Fixed 32k	0.78	64–64	3/3
FreeSolv	Joint 2–32k	1.02	64–128	3/3
BACE	Fixed 2k	1.84	512–512	0/3
BACE	Fixed 32k	0.72	32–32	3/3
BACE	Joint 2–32k	0.98	32–32	3/3
QM8	Fixed 2k	857.95	512–512	0/3
QM8	Fixed 32k	111.75	128–128	3/3
QM8	Joint 2–32k	451.20	128–256	3/3

5 Native baseline constraints and resource profiles

DataSAIL's native group mapping is observed immediately before its unchanged solver call, after its own overflow handling. Both matched controls preserve all rows and exact native counts; one also preserves every observed group. This does not equate linked cluster similarities with sums of entity-level similarities. Reported native infeasibility concerns the declared clustering, capacity and solver configuration, not DataSAIL's ability to process the dataset

under every possible setting.

Table 3: Global overlap relative to count-matched random. Both low-rank controls use exact native counts. The Groups control also preserves every group passed to DataSAIL's solver. Equal groups do not imply identical coarsened and entity-level objectives. One native solve is recorded per configuration; these configurations are not independent datasets.

Dataset	K	Budget (s)	DataSAIL	Counts	Groups
BACE	75	300	0.828	0.750	0.767
BACE	75	60	0.851	0.750	0.767
BACE	30	300	0.767	0.750	0.779
BACE	30	60	0.767	0.750	0.779
BBBP	9	300	0.514	0.458	0.514
BBBP	9	60	0.514	0.458	0.514
BBBP	101	300	0.916	0.703	0.743
BBBP	101	60	0.916	0.703	0.743
BBBP	40	300	0.745	0.692	0.749
BBBP	40	60	0.745	0.692	0.749
ESOL	56	300	0.597	0.524	0.652
ESOL	56	60	0.607	0.524	0.652
ESOL	22	300	0.648	0.524	0.664
ESOL	22	60	0.648	0.524	0.664
FreeSolv	32	300	0.510	0.415	0.574
FreeSolv	32	60	0.510	0.415	0.574

Table 4: Mean maximum test-to-training similarity. Both low-rank controls use exact native counts. The Groups control also preserves every group passed to DataSAIL's solver. Equal groups do not imply identical coarsened and entity-level objectives. One native solve is recorded per configuration; these configurations are not independent datasets.

Dataset	K	Budget (s)	DataSAIL	Counts	Groups
BACE	75	300	0.459	0.525	0.447
BACE	75	60	0.485	0.525	0.447
BACE	30	300	0.366	0.525	0.303
BACE	30	60	0.366	0.525	0.303
BBBP	9	300	0.135	0.170	0.135
BBBP	9	60	0.135	0.170	0.135
BBBP	101	300	0.301	0.252	0.237
BBBP	101	60	0.301	0.252	0.237
BBBP	40	300	0.235	0.246	0.239
BBBP	40	60	0.235	0.246	0.239

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Table 4 (continued)

Dataset	K	Budget (s)	DataSAIL	Counts	Groups
ESOL	56	300	0.296	0.270	0.260
ESOL	56	60	0.265	0.270	0.260
ESOL	22	300	0.273	0.270	0.286
ESOL	22	60	0.273	0.270	0.286
FreeSolv	32	300	0.178	0.368	0.285
FreeSolv	32	60	0.178	0.368	0.285

Ordinary graph construction forms the union of positive top-k pairwise similarities, with one undirected edge per unordered pair. Integer scaling respects a conservative 32-bit accumulation budget, and any count repair uses the same integer-weight graph. Native integer cuts are independently checked against edge sums. Both native and repaired partitions are saved.

The primary grid times complete native split calls with two pinned cores. Feature loading, imports and quality scoring are excluded and warm-up is recorded separately. The memory-profile repetition captures peak process RSS before the first scoring call; DataSAIL's later matched controls cannot inflate its reported native memory. This scope includes input arrays, imports and the common small low-rank warm-up. Different time-limited native solves can return different assignments; each profile retains its own outputs.

Table 5: Fresh-process native memory profiles at seed zero. Memory is captured before the first scoring call or any matched controls and includes framework/input costs and the common small warm-up. Graph configuration is degree; DataSAIL configuration is cluster policy/solver seconds. Native profiling repeats can differ from a time-limited earlier solve; their own assignments and quality scores are retained. Solver-reported infeasibility has no imputed timing or memory.

Dataset	Method	Configuration	Call (s)	RSS (GiB)
BACE	Combined	–	0.56	0.579
BACE	DataSAIL	coarse/60	–	Infeasible
BACE	DataSAIL	coarse/300	–	Infeasible
BACE	DataSAIL	fine/60	71.70	1.051
BACE	DataSAIL	fine/300	312.13	1.054
BACE	DataSAIL	medium/60	37.12	0.841
BACE	DataSAIL	medium/300	31.29	0.842
BACE	Graph	16	0.51	0.617
BACE	Graph	32	1.13	0.619
BACE	Graph	128	2.30	0.647
BACE	Plain	–	0.39	0.558
BBBP	Combined	–	0.70	0.648
BBBP	DataSAIL	coarse/60	37.86	0.865

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Table 5 (continued)

Dataset	Method	Configuration	Call (s)	RSS (GiB)
BBBP	DataSAIL	coarse/300	28.34	0.866
BBBP	DataSAIL	fine/60	95.65	1.307
BBBP	DataSAIL	fine/300	344.22	1.301
BBBP	DataSAIL	medium/60	42.45	0.921
BBBP	DataSAIL	medium/300	42.86	0.921
BBBP	Graph	16	0.62	0.628
BBBP	Graph	32	0.70	0.628
BBBP	Graph	128	8.40	0.664
BBBP	Plain	–	0.45	0.573
ESOL	Combined	–	0.43	0.559
ESOL	DataSAIL	coarse/60	–	Infeasible
ESOL	DataSAIL	coarse/300	–	Infeasible
ESOL	DataSAIL	fine/60	71.94	0.954
ESOL	DataSAIL	fine/300	90.20	0.969
ESOL	DataSAIL	medium/60	4.94	0.792
ESOL	DataSAIL	medium/300	5.05	0.790
ESOL	Graph	16	0.50	0.613
ESOL	Graph	32	0.56	0.618
ESOL	Graph	128	10.73	0.638
ESOL	Plain	–	0.27	0.546
FreeSolv	Combined	–	0.27	0.535
FreeSolv	DataSAIL	coarse/60	–	Infeasible
FreeSolv	DataSAIL	coarse/300	–	Infeasible
FreeSolv	DataSAIL	fine/60	4.97	0.801
FreeSolv	DataSAIL	fine/300	4.94	0.800
FreeSolv	DataSAIL	medium/60	–	Infeasible
FreeSolv	DataSAIL	medium/300	–	Infeasible
FreeSolv	Graph	16	0.50	0.604
FreeSolv	Graph	32	0.44	0.606
FreeSolv	Graph	128	0.68	0.623
FreeSolv	Plain	–	0.20	0.533
Lipophilicity	Combined	–	13.18	0.782
Lipophilicity	Graph	16	3.25	0.646
Lipophilicity	Graph	32	2.99	0.655
Lipophilicity	Graph	128	8.11	0.723
Lipophilicity	Plain	–	11.01	0.608
Tox21	Combined	–	4.10	1.026
Tox21	Graph	16	6.93	0.678
Tox21	Graph	32	2.92	0.694
Tox21	Graph	128	7.98	0.813
Tox21	Plain	–	1.53	0.653

6 Scaling, quality audits and censored outcomes

The current CPU protocol uses nested ChEMBL samples, 2048-bit fingerprints, rank 256, four restarts, two refined candidates, and exact or 1 percent capacity corridors. The plain setting is attempted through two million entities. The exact-neighbour combined CPU setting has a separate 10,000-entity range. Each timed split follows a 100-entity warm-up. A 1,800-second end-to-end case budget includes loading and independent audits; its expiration does not imply that the split call alone consumed that time. Partial outputs and terminal failure records remain preserved.

For completed calls, the deterministic factor is reconstructed outside the split clock and its landmark indices and factor objective checked against the public split diagnostics. Independent pair samples evaluate the returned cut. Up to 128 uniform test queries are compared with the complete training fold. The nearest-similarity mean uses a conservative bounded-mean sampling interval. Input arrays and framework overhead are included in process RSS; peak RSS immediately after splitting is distinguished from the later process peak.

Table 6: Current CPU scaling. Two-core cases run on lambda2; 32-core cases run on lambda13 and are a separate hardware protocol. Call time excludes feature loading, warm-up and independent scoring. RSS is the fresh-process peak immediately after splitting, including loaded input/features and framework overhead. Cut-error intervals concern the returned partition and use total powered-kernel mass. A case-budget outcome supplies no imputed split-call time or memory.

Cores/setting	Entities	Tolerance	Seed	Call (s)	RSS (GiB)	Cut-error interval
2/Plain	10,000	0.0	0	6.8	1.20	[0.0034, 0.0075]
2/Plain	10,000	0.01	0	4.4	1.20	[0.0034, 0.0075]
2/Plain	100,000	0.0	0	508.3	4.09	[0.0037, 0.0079]
2/Plain	100,000	0.0	1	567.2	4.09	[0.0030, 0.0071]
2/Plain	100,000	0.0	2	843.6	4.09	[0.0029, 0.0070]
2/Plain	100,000	0.01	0	50.7	4.09	[0.0037, 0.0078]
2/Plain	100,000	0.01	1	40.1	4.09	[0.0029, 0.0071]
2/Plain	100,000	0.01	2	39.1	4.09	[0.0028, 0.0070]
2/Plain	1,000,000	0.0	0	–	–	Case budget
2/Plain	1,000,000	0.01	0	–	–	Case budget
2/Plain	2,000,000	0.0	0	–	–	Case budget
2/Plain	2,000,000	0.0	1	–	–	Case budget
2/Plain	2,000,000	0.0	2	–	–	Case budget
2/Plain	2,000,000	0.01	0	–	–	Case budget
2/Plain	2,000,000	0.01	1	–	–	Case budget
2/Plain	2,000,000	0.01	2	–	–	Case budget
2/Combined	10,000	0.0	0	8.7	1.26	[0.0180, 0.2173]
2/Combined	10,000	0.01	0	10.4	1.28	[0.0167, 0.2159]
32/Plain	2,000,000	0.01	0	250.3	58.83	[0.0035, 0.0077]

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Table 6 (continued)

Cores/setting	Entities	Tolerance	Seed	Call (s)	RSS (GiB)	Cut-error interval
32/Plain	2,000,000	0.01	1	243.9	58.83	[0.0036, 0.0078]
32/Plain	2,000,000	0.01	2	234.9	58.83	[0.0026, 0.0068]

Earlier-release H100 timings appear only in the archival section. They are not measurements of the current code, and hardware/version differences are not interpreted as algorithmic speedups.

The separate largest-scale current-code protocol runs on lambda13 with two disjoint 32-core CPU allocations and hidden CUDA devices. Three two-million-entity plain cases use the same default numerical package and a 1 percent corridor. Packed inputs are prepared once in an immutable node-local cache. A completed timed split is persisted before a separate independent-audit phase, so an interrupted audit need not repeat optimization. The phase budget is 3,600 seconds. All three splits and their audits complete. Their timings are not placed on the two-core curve or used to infer an algorithmic speedup between hosts.

7 Synthetic independent-cohort protocol

Each generated instance contains fifty observed families with forty entities each, four Gaussian stable features, Gaussian family effects with standard deviation two, and observation noise with standard deviation 0.25. The stable response is $2z_1 - z_2 + 0.5z_3$. Full features concatenate a fixed 100-family one-hot universe with $0.05(\mathrm{clip}(z, -4, 4) + 4)$. Only the first fifty families occur in the split pool. Known-family and new-family evaluation observations use independent random draws; the latter use the remaining fifty independent family effects. A declared shared universe of identity features supplies no response information about unseen families.

The invariant ridge reference receives the unmodified stable features and uses penalty one. Five-neighbour regression uses the full representation. Both recipes are fixed. The invariant reference's access to generative features is privileged and is part of the mechanism design, not an asserted real-data feature-selection ability. The plain cosine case uses an exact finite feature map when available, while powering uses Nyström features. The family-reference split assigns entire generator families and has the same 1600/200/200 counts. Labels never enter structural split construction.

For each trained model, interpolation risk is evaluated on independently drawn observations from families present in that model's training fold. New-family risk uses the same independent unseen-family cohort for all policies within an instance. Different splits consequently need not share an identical interpolation cohort. The target is the declared conditional transfer task, not an unconditional comparison of model accuracy between split policies. Twenty seeds define twenty independent generated instances. Per-instance results, ranking comparisons and assignment hashes are retained.

8 Objective identities and conditional bounds

Let $H \in \{0,1\}^{n\times k}$ be the assignment matrix, with one nonzero per row, and let h_c denote its columns. The cross-fold mask is $\mathbf{1}\mathbf{1}^\top - HH^\top$ and has zero diagonal. Consequently,

$$\mathcal{C}_S(H) = \tfrac{1}{2}\left(\mathbf{1}^\top S\mathbf{1} - \sum_c h_c^\top S h_c\right). \tag{1}$$

Substituting $S = BB^\top$ yields Equation 3. Group contraction is exact because a whole group's contribution to every block sum is the sum of its rows. This statement concerns retained groups; dissolving a group changes the admissible partition set rather than invalidating the identity.

For a positive-semidefinite kernel with features ϕ_i, its integer elementwise power has features $\phi_i^{\otimes p}$ because $\langle \phi_i^{\otimes p}, \phi_j^{\otimes p}\rangle = \langle \phi_i, \phi_j\rangle^p$. This gives the stated positive-semidefinite property directly and explains why arbitrary fractional powers are not accepted.

For nonnegative similarities, let $N_\tau(z)$ count unordered crossing pairs with $S_{ij} \geq \tau > 0$. Each such pair contributes at least τ^p, so

$$\mathcal{C}_{S^{\circ p}}(z) \geq \tau^p N_\tau(z). \tag{2}$$

Thus the powered cut supplies a soft penalty on the number of strong crossing pairs. For a fixed nonempty set of crossing pairs, $\lim_{p\to\infty} \mathcal{C}_{S^{\circ p}}(z)^{1/p} = \max_{i<j:z_i\neq z_j} S_{ij}$, by the finite-dimensional ℓ_p norm limit. This is the maximum over all crossing pairs, not the average of per-test-point nearest neighbours used in NNL. The power therefore provides a principled tail surrogate, without making the optimized objective identical to the evaluation metric. A rank-limited factor introduces a separate approximation error for each power.

For symmetric residual $E = S - BB^\top$, the spectral norm inequality $|v^\top E v| \leq \|E\|_2\|v\|^2$ gives

$$|\mathcal{C}_S(H) - \mathcal{C}_{BB^\top}(H)| \leq \tfrac{1}{2}\|E\|_2\left(\|\mathbf{1}\|^2 + \sum_c \|h_c\|^2\right) \tag{3}$$

$$= n\|E\|_2 \equiv \delta. \tag{4}$$

The bound is uniform across assignments and can be loose in practice. If a solver returns $\widehat{H}$ with approximate-kernel objective at most η above the minimum over a specified feasible set, then

$$\mathcal{C}_S(\widehat{H}) \leq \min_{H \text{ feasible}} \mathcal{C}_S(H) + 2\delta + \eta. \tag{5}$$

Our heuristic does not certify η, so this conditional inequality is not an empirical optimality guarantee. It also requires comparing the same feasible set; grouping, count changes and kernel sharpening alter the optimization problem.

Moving a factor row b from block a to c replaces (p_a, p_c) by $(p_a - b, p_c + b)$. Expansion of Equation 3 gives $\Delta\mathcal{C} = b^\top p_a - b^\top p_c - \|b\|^2$. For a swap of factor rows u and v, the reduction equals the sum of their single-move reductions minus $2u^\top v$. These identities hold for group sums as well. Searching only a finite set of swap candidates preserves the validity of an accepted move, but does not establish that no improving swap exists elsewhere.

If all fold counts equal a fixed value m, kernel k-means maximizes $m^{-1}\sum_c \|p_c\|^2$ and agrees with dense-cut minimization. For unequal fixed counts it maximizes a differently weighted sum, and for a capacity corridor the weights themselves change with the assignment. This is why we do not transfer ordinary kernel k-means convergence or approximation guarantees to the present solver without further assumptions.

9 ODAC25 metadata-only composition case study

9.1 Source, transfer and original-split reconciliation

This descriptive application uses the filtered ODAC25 release [Sriram et al., 2026] through the ColabFit conversion [ColabFit, 2026]. The original data documentation is retained with its Hugging Face revision. The pinned ColabFit revisions are:

Training: `064de0077b3de4745f58c58e7a34da933c8b727e`
Validation: `99a348267a440fbdb45725c7e790eb454c4ad796`

We read the columns `names`, `chemical_formula_hill` and `multiplicity` from all 82 configuration shards. HTTPS byte-range requests retrieve Parquet footers and the selected column chunks; the reader rejects a non-206 response or a mismatched Content-Range before reading the body. It never downloads a complete shard or the original ASE-LMDB archive. The retained ledger records every pinned source URL, selected columns, source file size, payload bytes, range-request count and compact-cache checksum.

The metered production payload is 63,431,987 bytes. The source shards occupy 200.5 GB in total (200,522,544,000 bytes). A recorded pilot transfers 248,514 bytes; another small failed projected-column pilot and documentation/API requests are not fully metered. Thus 63.4 MB is the production payload, not an exact whole-session network counter. The expanded compact JSON caches occupy about 1.2 GiB locally and are not included in that transfer figure. Those caches can be regenerated through the pinned metadata requests. The paper builds from retained compact results and assignments without network access or trajectory data.

The mirror contains 37,456,852 configuration rows. Multiplicities restore the original counts: 36,920,036 training configurations and 791,638 validation configurations, totalling 37,711,674. Each compact record retains its original source fold. Counts reconcile exactly with the official filtered archives, but this is not a row-by-row verification of their atomic structures or energy corrections. The new two-fold assignments pool both original folds. No original test archive is listed in the documentation used here.

9.2 Identity rules and composition representatives

The mirror retains trajectory names but not the official `atoms.info.mof_name` field. We therefore identify conservative *inferred families*, not certified original MOF identities. Adsorption and GCMC suffixes are removed; synthetic `Gen<number>` names are grouped; six-letter CSD parents combine polymorph, defect and functionalization variants; other source prefixes retain the documented parsing rule. Connected components also link shared configuration aliases, identical reduced observed formulas, and identical known reduced bare-host formulas. These rules give 2,913 inferred families and 2,050 immutable components. No component is dissolved. Shared composition does not establish shared topology, and the grouping rule may deliberately combine distinct structures.

Bare-host formulas come from bare-MOF rows or subtraction of explicit adsorbate stoichiometries from recognized trajectory names. The latter agrees with 97,870 corresponding

bare trajectories with no formula mismatch and recovers formulas for 423 additional inferred families. The 92 remaining families have no resolved bare formula; none is guessed. All optimization features consistently use the *observed full-system* composition, including adsorbates. Known-host formulas add identity links only.

For each of 205,269 distinct reduced observed formulas, let x_f be its 118-dimensional elemental atom-fraction vector. Each component g receives the unweighted mean of its distinct x_f vectors. The optimizer's bounded Euclidean kernel uses standardization fitted over these component representatives. This aggregation is an additional representation choice: the kernel of a mean feature vector generally differs from the mean of the underlying kernels. It is not claimed to preserve the original configuration- level cut exactly.

9.3 Weighted optimization and matched controls

This case freezes the original v0.1 numerical factor source, included under `odac25_case/method_source`, separately from the primary v0.2.0.dev3 experiments. A rank-256 Nyström factor supplies component rows b_g. If w_g is the component's configuration multiplicity, the fold sum is $p_c = \sum_{g:z_g=c} w_g b_g$, and the two-fold surrogate cut is $p_{\text{train}}^{\top} p_{\text{val}}$. Binary linear assignment enforces the exact integer configuration counts. Starting from seeded random linear costs, each proposal maximizes self-excluded within-fold scores; a proposal is accepted only when its weighted factor cut decreases. At most fifteen proposals are allowed. Retained solver logs distinguish a valid incumbent from a certified linear optimum; neither establishes global optimality of the quadratic partition objective.

The control before optimization is the adapter's exact feasible initialization. It is constructed without factor values using random linear costs and the same count constraint; it is not a uniform sampler of feasible group partitions. The dense control uses the same initialization, whole groups, counts and guarded proposal heuristic with the full component kernel. It is competitive at this size, and no low-rank speedup is inferred from this case. Optimization-only timings are retained with their scope; they exclude metadata aggregation, feature construction, factorization and scoring. The earlier 80/10/10 split balanced component counts and is not used in this count-matched comparison.

9.4 Evaluation on recorded compositions and uncertainty

Evaluation returns to the 205,269 recorded formula vectors, rather than scoring only component means. Standardization is fitted once to the complete configuration-weighted formula pool, using the same mean and sample variance for all assignments. Thus the scoring reference differs explicitly from the component-level optimization standardization. Zero-variance dimensions are omitted. With configuration counts N, n_T and n_V, the global fraction is estimated from independently sampled total-pool and cross-fold pairs; actual self-pair expectation $1/N$ is removed from the total-pool denominator. The ratio is normalized by $2n_T n_V/[N(N-1)]$. Each pair mean uses 200,000 draws. Nearest-training similarity averages 512 configuration-weighted validation queries, each compared with *every* distinct training formula.

The reported Monte Carlo standard errors describe pair/query sampling only. They are not confidence intervals over independent frameworks, datasets or model initializations.

Three optimizer seeds produce only 2 distinct low-rank assignments: seeds zero and two are identical, and seed one changes nine component labels. The validation fold contains 61–64 components, with the two largest accounting for 40.6% of its configuration weight.

Method	Seed	Global ratio	Mean nearest	NN MC SE
Original	–	0.9692	0.8464	0.0104
Groups before optimization	0	0.8726	0.5743	0.0109
Groups before optimization	1	0.8318	0.5589	0.0111
Groups before optimization	2	0.8248	0.5256	0.0119
Low-rank optimization	0	0.2829	0.1937	0.0060
Low-rank optimization	1	0.2833	0.1957	0.0060
Low-rank optimization	2	0.2829	0.1937	0.0060
Dense-kernel control	0	0.2868	0.1644	0.0051
Dense-kernel control	1	0.2872	0.1579	0.0047
Dense-kernel control	2	0.2866	0.1538	0.0046

Table 7: ODAC25 individual observations. MC SE describes configuration-query sampling error only. It does not describe uncertainty over independent MOFs, projects, or prediction models.

Independent float64 SciPy nearest-neighbour calculations for 32 additional sampled validation queries per arm agree with Torch within 0.00007 similarity. The largest returned low-rank cut discrepancy against the exact component kernel is 1.92% of that cut; this denominator differs from the total-kernel-mass normalization in the primary fidelity study. Checksum verification covers saved inputs, producer scripts, the frozen factor implementation and results. Small exhaustive tests separately check exact integer assignment and agreement between weighted and explicitly expanded configuration objectives.

9.5 Scientific scope

For seed zero, the total variation between mean training and validation atomic fractions is 0.339, compared with 0.031 for the original split. Phosphorus presence changes from 1.6% in training to 36.8% in validation. Under the original split, 46.0% of validation configuration weight shares a reduced observed formula with training. Its absence in the new split follows from the grouping contract and must not be attributed to optimization alone. Original composition overlap is also not proof that identical atomic geometries cross folds. This case establishes a metadata-efficient composition-shift holdout, with improvements beyond the imposed grouping constraints. It evaluates no energies, forces, prediction errors, target-distribution shifts or prospective project outcomes, and it is not a scalability test over millions of independent optimizer entities.

10 Additional current-version optimization controls

The added primary control grid contains 432 completed cases: 288 iteration-controlled dense/factor cases, 72 equal-clock cases, 18 component cases containing ten settings each, and 54 CCPart cases containing native and matched interventions where feasible. Separate additions provide 36 nested ChEMBL dense controls and 72 preempted wall-clock controls. Cases and within-case settings are not independent datasets. All use the unchanged primary numerical package; the control source, upstream Hestia source and complete input/assignment identities are frozen separately in `revision_20261103_controls`.

10.1 Original-cut error and true-kernel optimization quality

For a returned partition z with positive true cut, we calculate

$$\epsilon_{\mathrm{cut}}(z) = \frac{|\mathcal{C}_S(z) - \mathcal{C}_{BB^\top}(z)|}{\mathcal{C}_S(z)}.$$

For powered factors, both S and the denominator use the stated power. Absolute and kernel-mass-normalized errors remain available in the numerical records. A zero true cut has undefined cut-relative error, with an explicit status rather than an arbitrary denominator offset. All 900 earlier full-kernel comparisons have positive true cuts.

The maximum errors below characterize only tested assignments. A small cut-value error does not certify optimizer quality or bound any individual nearest-neighbour exposure. The exact small-instance enumeration in the preceding sections separately establishes approximation and optimization gaps over its finite feasible sets.

Table 8: Reanalysis of all 900 saved full-kernel fidelity comparisons: six datasets, three seeds, five ranks, two powers and five tested partitions. Entries are maxima. Mass and cut denominators use the same power as the factor. Returned cut restricts to the optimized partition; the other maxima include four count-preserving perturbations. All true cuts in this suite are positive; zero-cut relative errors would be undefined.

Power	Rank	Mass error (%)	Cut error (%)	Returned cut (%)
1	32	5.99	23.16	21.39
1	64	3.23	12.48	10.75
1	128	1.65	6.39	5.31
1	256	0.70	2.71	1.72
1	512	0.27	1.05	0.58
3	32	22.59	94.20	94.20
3	64	20.30	88.92	88.92
3	128	16.89	78.70	78.70
3	256	13.67	64.47	64.47
3	512	9.36	44.14	44.14

Dataset	n	Dense	Factor	Error (%)	Dense (s)	Factor (s)
FreeSolv	642	0.420	0.420	0.73	0.03	0.18
ESOL	1128	0.527	0.527	0.83	0.08	0.20
BACE	1513	0.752	0.752	0.14	0.12	0.25
BBBP	2039	0.694	0.695	1.03	0.28	0.30
Lipophilicity	4200	0.795	0.795	1.03	1.59	0.65
Tox21	7823	0.567	0.565	1.66	6.72	1.14

Table 9: Direct dense and rank-256 factor optimization with identical initial assignments, 25 assignment iterations at most, the same refinement limit, and a 1% size corridor. Dense and Factor are original-kernel global overlap relative to count-matched random. Error is factor cut error divided by the true cut at its returned assignment. Times include representation construction and optimization on two CPU cores, excluding loading, warm-up and independent scoring. Values average three seeds. These are heuristic solutions, not dense optima.

Table 10: Iteration-controlled dense comparisons over six datasets and three seeds. Ratios are averaged within datasets before averaging datasets. Cut / dense compares original powered-kernel cuts of the separately optimized partitions; a value below one can reflect heuristic search trajectories and does not prove approximation superiority. Max error is the largest cut-relative representation error across returned partitions. Storage counts the float32 kernel or factor array only, not process RSS.

Power	ϵ	Rank	Cut / dense	Max error (%)	Time / dense	Storage / dense
1	0.00	64	1.004	11.33	0.846	0.042
1	0.00	256	0.999	1.72	1.862	0.169
1	0.00	512	1.000	0.62	3.452	0.338
1	0.01	64	1.004	11.47	0.710	0.042
1	0.01	256	1.000	1.73	2.127	0.169
1	0.01	512	1.000	0.62	3.975	0.338
3	0.00	64	1.106	90.73	0.595	0.042
3	0.00	256	1.028	66.81	1.637	0.169
3	0.00	512	1.003	46.18	2.870	0.338
3	0.01	64	1.078	90.75	0.860	0.042
3	0.01	256	1.025	67.02	2.009	0.169
3	0.01	512	1.001	46.18	3.995	0.338

Table 11: Complete nested ChEMBL dense/factor grid. Global is original-kernel overlap normalized by count-matched random; error is maximum returned-cut-relative approximation error over three seeds. Timings and peak RSS are means. Both representations use the same bounded assignment/refinement protocol, with no response-label information.

Entities	Arm	Rank	Seconds	RSS GiB	Global	Max error (%)
4,000	dense	–	1.38	0.92	0.791	–
4,000	factor	64	0.40	0.80	0.794	9.51
4,000	factor	256	1.08	0.82	0.791	1.96
4,000	factor	512	2.11	0.87	0.791	0.67
16,000	dense	–	23.79	3.93	0.784	–
16,000	factor	64	1.48	1.36	0.787	10.14
16,000	factor	256	4.52	1.36	0.785	2.42
16,000	factor	512	8.34	1.36	0.784	0.97
32,000	dense	–	82.68	12.04	0.783	–
32,000	factor	64	3.08	1.83	0.787	10.69
32,000	factor	256	8.74	1.83	0.784	2.16
32,000	factor	512	16.01	1.83	0.783	0.85

10.2 Time, memory and equal-clock controls

The dense oracle stores the kernel directly, without a full-kernel eigendecomposition. Its float32 operations supply the same interface as the factor to the unchanged assignment and refinement routines. Final original-kernel scoring is independent and uses float64 division and summation. Native peak RSS is captured before scoring and includes input and framework costs. The array column measures just the representation, so these memory quantities must not be interchanged.

The direct oracle sums crossing kernel blocks with indexed array operations, which allocate temporary arrays. Its peak RSS measures this implementation, including those temporaries; it is not a lower bound on memory for all possible dense optimizers. The cost comparison likewise measures the declared implementations and budgets rather than an implementation-independent speedup.

Table 12: Fresh-process profiles for the main dense control. RSS is recorded before any independent scoring, includes imports, inputs and identical warm-up, and is not an incremental representation-memory measurement. Array bytes are reported separately. Mean of three seeds.

Dataset	Arm	RSS (GiB)	Array (MiB)	Build (s)	Total (s)
FreeSolv	dense	0.511	1.57	0.01	0.03
FreeSolv	factor	0.530	0.63	0.16	0.18
ESOL	dense	0.525	4.85	0.03	0.08
ESOL	factor	0.543	1.10	0.18	0.20

Dataset	Arm	RSS (GiB)	Array (MiB)	Build (s)	Total (s)
BACE	dense	0.544	8.73	0.05	0.12
BACE	factor	0.549	1.48	0.22	0.25
BBBP	dense	0.568	15.86	0.11	0.28
BBBP	factor	0.566	1.99	0.27	0.30
Lipophilicity	dense	0.721	67.29	0.47	1.59
Lipophilicity	factor	0.592	4.10	0.59	0.65
Tox21	dense	1.181	233.46	1.45	6.72
Tox21	factor	0.615	7.64	1.07	1.14

The primary equal-time protocol runs construction and search in a timed subprocess. An external supervisor stops that process at the deadline and scores its last serialized feasible checkpoint completed before the cutoff. Scheduling and termination can finish slightly after that cutoff, but late work cannot improve the selected partition. The largest measured termination overshoot is 0.114 seconds.

Table 13: Equal-time controls with an externally stopped optimizer. Representation construction and search share the budget; only a feasible checkpoint whose evaluation and serialization finished before the deadline is scored. Late work cannot improve the selected partition. Times include scheduling and termination overshoot; scores and times average three seeds. The initial feasible partition is available before timing starts.

Dataset	Budget (s)	Dense	Factor	Dense (s)	Factor (s)
FreeSolv	1	0.420	0.420	1.064	1.064
FreeSolv	5	0.419	0.420	5.064	5.064
ESOL	1	0.526	0.526	1.064	1.064
ESOL	5	0.526	0.526	5.064	5.064
BACE	1	0.750	0.750	1.064	1.064
BACE	5	0.750	0.750	5.064	5.064
BBBP	1	0.693	0.693	1.064	1.064
BBBP	5	0.693	0.693	5.064	5.064
Lipophilicity	1	0.797	0.795	1.064	1.064
Lipophilicity	5	0.794	0.794	5.064	5.064
Tox21	1	1.002	0.857	1.080	1.064
Tox21	5	0.569	0.564	5.064	5.064

An earlier cooperative sensitivity protocol checks deadlines between single-step calls to the same assignment/refinement routines and uses the same ordered restart seeds. Construction is charged to the budget, and an operation already started is allowed to finish. Its largest observed overrun is 0.99 seconds. Actual work counts, construction times and construction-exhausted outcomes are retained per case. This earlier protocol cannot be interpreted as a hard real-time resource comparison.

Table 14: Equal-clock controls including representation construction. Both arms repeatedly call the current one-step assignment and refinement routines, using the same restart-seed sequence. Clocks are checked between operations: the last operation may overrun, and the table reports maximum overrun in seconds. If construction exhausts the budget, the unchanged feasible initial partition is returned. Global scores average three seeds. These cooperative budgets are not hard real-time deadlines.

Dataset	Budget (s)	Dense	Factor	Dense overrun	Factor overrun
FreeSolv	1	0.420	0.420	0.002	0.003
FreeSolv	5	0.419	0.420	0.005	0.002
ESOL	1	0.526	0.526	0.006	0.004
ESOL	5	0.526	0.526	0.009	0.003
BACE	1	0.750	0.750	0.011	0.002
BACE	5	0.750	0.750	0.010	0.003
BBBP	1	0.693	0.693	0.040	0.004
BBBP	5	0.693	0.693	0.015	0.005
Lipophilicity	1	0.796	0.795	0.217	0.008
Lipophilicity	5	0.794	0.794	0.234	0.006
Tox21	1	1.002	1.002	0.995	0.282
Tox21	5	0.567	0.564	0.533	0.018

10.3 Current-version component controls

Omitting assignment or refinement leaves a feasible initial partition in place, rather than relaxing the capacity constraint. Four-restart controls refine two candidates; group controls use one warm start, one additional restart and one refined candidate. Timing the optimizer with reused factors isolates component cost, while construction costs remain explicit per case. The three grouped rows share their groups and feasible packing. Comparisons between grouped and ungrouped rows change the admissible partition set and are interpreted accordingly.

Table 15: Current-version component controls at rank 256 and a 1% size corridor. Each cell averages three seeds within each of six datasets, then datasets equally. The first seven rows are ungrouped; the last three share the same strictly preserved capped groups and feasible group initialization. Grouping changes the feasible set. Optimization time excludes shared factor and group construction, which are retained per case. No response labels enter these controls. Restart budgets and group starts are specified in Methods.

Setting	Global	Nearest	Optimize (s)
Feasible initialization	1.001	0.622	0.00
Assignment only	0.626	0.346	0.04
Refinement only	0.998	0.621	0.03
Assignment + refinement	0.626	0.346	0.07

Setting	Global	Nearest	Optimize (s)
Four restarts / two refined	0.625	0.369	0.18
Uniform landmarks / four starts	0.625	0.364	0.14
Cubed factor / four starts	0.669	0.326	0.14
Fixed groups / feasible packing	0.956	0.388	0.00
Fixed groups / ordinary factor	0.792	0.321	0.09
Fixed groups / cubed factor	0.748	0.292	0.11

10.4 CCPart native sizes and group-preserving interventions

The native CCPart result is produced by the pinned upstream implementation, not by a reimplementation of component packing. Threshold similarities are exhaustively constructed, without response labels, and construction belongs to its native runtime. Every entity is retained. The unmodified packing routine need not realize the exact requested counts or the low-rank corridor. Three deterministic input permutations characterize order sensitivity.

The count-matched low-rank intervention may split components. The second intervention preserves all native components and their exact realized counts; it therefore retains the original threshold separation. Both start from the native assignment. Timings for the interventions reuse components and must not be presented as complete standalone pipelines with free group construction. Native memory is recorded before independent scoring or matched optimization.

Table 16: CCPart retains every entity. Count ranges span three input permutations. Group count and largest component show the native threshold contract. Whole components are retained; fold sizes are not silently repaired to the low-rank corridor.

Dataset	τ	n	Train	Val	Test	Groups	Largest
FreeSolv	0.4	642	515	63	64	134	427
FreeSolv	0.6	642	514	64	64	467	31
FreeSolv	0.8	642	514	64	64	586	6
ESOL	0.4	1128	904	112	112	245	625
ESOL	0.6	1128	904	112	112	722	35
ESOL	0.8	1128	904	112	112	1024	11
BACE	0.4	1513	1286	94	133	29	650
BACE	0.6	1513	1226	140	147	120	308
BACE	0.8	1513	1211	151	151	776	69
BBBP	0.4	2039	1634	202	203	539	441
BBBP	0.6	2039	1633	203	203	1177	135
BBBP	0.8	2039	1633	203	203	1820	7
Lipophilicity	0.4	4200	3362	419	419	830	2169
Lipophilicity	0.6	4200	3362	419	419	1788	136
Lipophilicity	0.8	4200	3362	419	419	3366	23
Tox21	0.4	7823	6262	779	782	1348	5444

Dataset	τ	n	Train	Val	Test	Groups	Largest
Tox21	0.6	7823	6259	782	782	4660	697
Tox21	0.8	7823	6259	782	782	7042	16

Table 17: Native CCPart quality and complete split-call cost, including exhaustive threshold similarities, connected components and packing. Memory is fresh-process peak RSS before scoring or matched interventions. Each row averages three input-order repetitions; all entities are retained.

Dataset	τ	Global	Nearest	Seconds	RSS (GiB)
FreeSolv	0.4	0.722	0.282	0.04	0.684
FreeSolv	0.6	0.923	0.434	0.04	0.684
FreeSolv	0.8	0.986	0.515	0.04	0.681
ESOL	0.4	0.819	0.289	0.08	0.693
ESOL	0.6	0.934	0.432	0.08	0.692
ESOL	0.8	0.978	0.521	0.09	0.691
BACE	0.4	0.768	0.265	0.13	0.706
BACE	0.6	0.832	0.375	0.11	0.704
BACE	0.8	0.936	0.689	0.12	0.699
BBBP	0.4	0.900	0.295	0.26	0.714
BBBP	0.6	0.954	0.404	0.18	0.712
BBBP	0.8	0.994	0.539	0.19	0.710
Lipophilicity	0.4	0.933	0.315	0.88	0.761
Lipophilicity	0.6	0.945	0.395	0.96	0.761
Lipophilicity	0.8	0.983	0.561	0.84	0.760
Tox21	0.4	0.866	0.304	2.49	0.839
Tox21	0.6	0.955	0.442	2.35	0.833
Tox21	0.8	0.999	0.535	2.70	0.833

Full per-seed quality, costs, native groups, assignments and checksums are included in the control records. An independent RDKit replay checks all original-kernel scores and verifies the absence of crossing native threshold edges against every original pair, as well as count and group preservation.

11 Metadata-only large molecular datasets with native partitions

PCQM4Mv2, MOSES and ogbg-molpcba extend the application evidence beyond ODAC25 and the nested ChEMBL scaling samples. The metered archive/CSV payload is 90.3 MB, including ZIP catalog ranges; small discovery pages, source JSON and HTTP headers are outside this tally. Retrieval is capped at 200 MiB of dataset payload and 128 MiB per

object. Only the required SMILES, mapping and native-index ZIP members are fetched. No coordinates, SDFs, trajectory archives, processed graph arrays or model checkpoints are downloaded.

MOSES files are pinned to an immutable maintainer commit and checked against their Git-LFS SHA-256 identities. Archive-member CRCs, extracted-file hashes, range requests and response sizes are retained for OGB sources. Raw metadata and locally generated packed fingerprints are reproducible caches outside the public evidence bundle. The CSV property and assay columns are not loaded for optimization. Sources and recipes are in `revision_20261104_native_splits`.

11.1 Native roles, identities and parsing

The original source assignments are checked for disjointness and complete row coverage before fingerprinting. PCQM4Mv2 retains its train, valid, test-dev and test-challenge roles; the released split is CID-based, as specified in the OGB documentation, rather than the originally intended scaffold split. MOSES retains train, random test and scaffold test. Ogbg-molpcba retains train, valid and test under its native scaffold assignment. No official held-out partitions are merged. Released PCQM4Mv2 test-dev and test-challenge response labels are unavailable. Its repartitioned assignments support structural and cost analysis; using them as a labelled prediction benchmark would require the corresponding responses.

The same radius-two, 1,024-bit Morgan fingerprint recipe is used throughout. RDKit 2026.03.5 cannot parse 39 PCQM4Mv2 records and two ogbg-molpcba records; their original positions and failure reasons are recorded. Every method uses the same retained rows and native count vector. All MOSES rows are retained. The canonical-identity audit uses hashes of isomeric canonical SMILES. Repeated records are retained, and neither native nor optimized assignments are silently deduplicated. In PCQM4Mv2, distinct source records can share a canonical graph; this does not establish identical atomistic geometries or a predictive effect.

Table 18: Metadata identity audit with the recorded RDKit version. Keys hash isomeric canonical SMILES; no chemical deduplication is silently applied. Crossing keys occur in more than one original native role. Parsing failures are fixed before all methods are compared, with original-row maps retained. Counts are molecular records, not atomistic frames.

Dataset	Original	Retained	Failed	Unique keys	Crossing keys
PCQM4Mv2	3,746,620	3,746,581	39	3,574,491	34,992
MOSES	1,936,962	1,936,962	0	1,936,962	0
ogbg-molpcba	437,929	437,927	2	437,927	0

The numerical package remains pinned to `f7e0866`. Its ordinary `SplitSpec` wrapper permits two or three roles. A separately hashed `NativeSplitSpec` subclass validates the four named PCQM4Mv2 roles and passes them to the unchanged k-way numerical routines. Exact/corridor small four-fold cases and invalid specifications are regression-tested. The initial API preflight failed during warm-up and is preserved separately; it produced no large optimized partitions and is not counted as a scientific outcome.

11.2 Generation cost and quality measurements

The production grid has 30 completed cases and 0 whole-case budget outcomes: one native, three count-matched random, three exact-count assignment-only and three corridor plain cases per dataset. Low-rank settings use 256 landmarks, four starts and at most 25 assignment iterations. Exact-count controls disable refinement; the 1% corridor setting refines two candidates when the unchanged size rule permits it. Above two million rows, point refinement is skipped. The two settings vary both feasibility and refinement work and do not isolate the effect of tolerance alone.

Fresh processes use CPUs 24–39 on lambda2, with CUDA hidden and native pools capped at 16. The measured public call includes its representation construction, assignment and enabled refinement. Imports, prepared-input loading, unpacking, input-adapter construction, common warm-up and independent auditing are outside that clock. Peak RSS is read before auditing and includes inputs/framework costs. Fingerprint preparation and metadata loading are recorded separately. Published native construction times and memory were not supplied, so native assignment loading is never presented as a competing optimization measurement. A 1,200-second whole-case cap includes every stage, and budget outcomes receive no imputed split-call time.

All methods share 262,144 uniformly sampled ordered pairs of distinct retained records per dataset. Original Tanimoto is evaluated exactly from packed bits. Global overlap is normalized by the analytic random crossing fraction at each returned count vector. Approximate 95% Monte Carlo intervals resample 64 independent pair blocks; they describe sampling uncertainty at fixed partitions, not future-dataset generalization or simultaneous coverage over every setting.

For every held-out role, 128 uniform queries without replacement are searched against the complete returned training fold. Exact packed-bit Tanimoto uses only mathematically valid upper-bound pruning; it is not approximate neighbour search. Per-query bootstrap intervals describe query-sampling uncertainty. The main-table aggregate weights role means by their actual populations, while the following table preserves their separate meanings. A similarity of one denotes fingerprint identity, not necessarily identical chemical graphs or geometries.

Table 19: Per-held-out-role exact nearest-training exposure on 128 uniform sampled queries per role and seed. Every query is compared with every returned training record. Fractions at one measure fingerprint identity, not identical geometries. Rows average completed seeds; full query-bootstrap intervals and per-seed observations accompany the evidence.

Dataset	Assignment	Role	Runs	Nearest	> 0.6	$= 1$
PCQM4Mv2	Native	valid	1	0.704	0.711	0.133
PCQM4Mv2	Native	test-dev	1	0.703	0.727	0.133
PCQM4Mv2	Native	test-challenge	1	0.722	0.797	0.141
PCQM4Mv2	Random	valid	3	0.760	0.878	0.146
PCQM4Mv2	Random	test-dev	3	0.771	0.904	0.154
PCQM4Mv2	Random	test-challenge	3	0.764	0.888	0.159

Dataset	Assignment	Role	Runs	Nearest	> 0.6	$= 1$
PCQM4Mv2	Exact assignment	valid	3	0.488	0.234	0.000
PCQM4Mv2	Exact assignment	test-dev	3	0.548	0.341	0.000
PCQM4Mv2	Exact assignment	test-challenge	3	0.552	0.375	0.000
PCQM4Mv2	1% corridor	valid	3	0.493	0.234	0.000
PCQM4Mv2	1% corridor	test-dev	3	0.554	0.328	0.000
PCQM4Mv2	1% corridor	test-challenge	3	0.540	0.320	0.000
MOSES	Native	test	1	0.739	0.930	0.008
MOSES	Native	test_scaffolds	1	0.707	0.883	0.000
MOSES	Random	test	3	0.759	0.938	0.016
MOSES	Random	test_scaffolds	3	0.755	0.945	0.010
MOSES	Exact assignment	test	3	0.608	0.487	0.000
MOSES	Exact assignment	test_scaffolds	3	0.601	0.510	0.000
MOSES	1% corridor	test	3	0.608	0.495	0.000
MOSES	1% corridor	test_scaffolds	3	0.587	0.424	0.000
ogbg-molpcba	Native	valid	1	0.652	0.672	0.023
ogbg-molpcba	Native	test	1	0.651	0.688	0.016
ogbg-molpcba	Random	valid	3	0.754	0.924	0.039
ogbg-molpcba	Random	test	3	0.761	0.922	0.039
ogbg-molpcba	Exact assignment	valid	3	0.563	0.427	0.000
ogbg-molpcba	Exact assignment	test	3	0.576	0.427	0.000
ogbg-molpcba	1% corridor	valid	3	0.559	0.409	0.000
ogbg-molpcba	1% corridor	test	3	0.568	0.414	0.000

Table 20: Complete native-split outcome inventory. Budget denotes the 1,200-second whole-case limit, including loading, warm-up, optimization and auditing; it is not an imputed split-call runtime. Native fixed-assignment construction times are unknown. Exact assignment explicitly disables refinement; the corridor setting uses the unchanged size-dependent refinement rule.

Dataset	Assignment	Seed	Outcome	Call (s)
PCQM4Mv2	Native	0	Complete	–
PCQM4Mv2	Random	0	Complete	0.1
PCQM4Mv2	Exact assignment	0	Complete	317.4
PCQM4Mv2	1% corridor	0	Complete	299.1
PCQM4Mv2	Random	1	Complete	0.1
PCQM4Mv2	Exact assignment	1	Complete	285.0
PCQM4Mv2	1% corridor	1	Complete	283.4
PCQM4Mv2	Random	2	Complete	0.1
PCQM4Mv2	Exact assignment	2	Complete	309.7
PCQM4Mv2	1% corridor	2	Complete	318.9
MOSES	Native	0	Complete	–

Dataset	Assignment	Seed	Outcome	Call (s)
MOSES	Random	0	Complete	0.1
MOSES	Exact assignment	0	Complete	185.3
MOSES	1% corridor	0	Complete	183.2
MOSES	Random	1	Complete	0.1
MOSES	Exact assignment	1	Complete	194.0
MOSES	1% corridor	1	Complete	195.7
MOSES	Random	2	Complete	0.1
MOSES	Exact assignment	2	Complete	178.5
MOSES	1% corridor	2	Complete	177.1
ogbg-molpcba	Native	0	Complete	–
ogbg-molpcba	Random	0	Complete	0.0
ogbg-molpcba	Exact assignment	0	Complete	38.5
ogbg-molpcba	1% corridor	0	Complete	42.0
ogbg-molpcba	Random	1	Complete	0.0
ogbg-molpcba	Exact assignment	1	Complete	39.6
ogbg-molpcba	1% corridor	1	Complete	36.4
ogbg-molpcba	Random	2	Complete	0.0
ogbg-molpcba	Exact assignment	2	Complete	43.5
ogbg-molpcba	1% corridor	2	Complete	42.3

11.3 Independent verification and returned-cut fidelity

Five regression tests check packed metrics against RDKit, zero-mass handling, four-role capacities and specification validation. Independent RDKit checks 1,024 sampled pair values per dataset. A second NumPy implementation verifies eight saved queries per held-out role against every training entity for native, exact-count and corridor seed-zero cases: 168 full-training query checks in total. Every saved assignment, count, query-selection sequence, threshold score, mean and sampling interval is also replayed from the saved artifacts.

The additional post-hoc fidelity diagnostic reconstructs sampled factor rows from saved landmark identities using the pinned double-construction and float32-storage formula. A regression against the actual producer factor checks the reconstruction. It evaluates both returned assignments for every seed without rebuilding a full factor or similarity matrix. No rank or method selection uses these outcomes. The absolute signed cut residual is normalized by original kernel mass and by the original returned cut. Approximate block-bootstrap intervals are reported separately from the earlier full-matrix tests and bounded accuracy-selection policy; they do not certify uniform fidelity or optimizer optimality.

Table 21: Post-hoc returned-cut fidelity at 256 landmarks. The factor is reconstructed only at uniform sampled pairs using saved landmarks and the pinned double-construction, float32-storage formula. Each cell is the largest estimate or interval endpoint over three seeds. The intervals are approximate Monte Carlo block-bootstrap intervals for the tested returned partition; they certify neither uniform fidelity nor optimizer optimality.

Dataset	Setting	Mass error (%)	Cut error (%)	Upper CI (%)
PCQM4Mv2	Exact assignment	0.40	3.50	3.71
PCQM4Mv2	1% corridor	0.39	3.50	3.71
MOSES	Exact assignment	0.32	1.30	1.38
MOSES	1% corridor	0.32	1.31	1.39
ogbg-molpcba	Exact assignment	0.41	1.53	1.62
ogbg-molpcba	1% corridor	0.41	1.55	1.64

Archival supporting studies: v0.1 review release

The following sections preserve protocols and results from the v0.1 review release. They provide broader representation, retention and predictive context. Their numerical kernel regularization, restart-selection order and some scoring normalizations differ from the current primary implementation and must not be read as its defaults. In particular, archival local ratios use their original ramp and random normalization; the main current comparison uses raw mean nearest-training similarity. Original source identities and per-study timing scopes remain with these observations. The archival external assays are not real chronological validation.

12 Dataset scope

Target slices follow the declared dataset-loader mapping and supply one scalar task per source for the prediction studies. Structural benchmarks optimize features without using responses, including entities whose target is missing. The study does not claim coverage of every target in Tox21, SIDER or MUV. MUV-466 has sparse labels and few actives; no predictive AUC is reported for MUV in this article. The dataset catalogue identifies the exact offline snapshots rather than silently substituting a newer database query.

Dataset	Entities	Labelled	Target	Positive
bace	1513	1513	Class	691
bbbp	2039	2039	p_np	1560
esol	1128	1128	measured log solubility in mols per litre	–
freesolv	642	642	expt	–
lipophilicity	4200	4200	exp	–
clintox	1480	1480	CT_TOX	112
hiv	41120	41120	HIV_active	1443
tox21	7823	7258	NR-AR	308
sider	1427	1427	Hepatobiliary disorders	743
muv	93087	14841	MUV-466	27
qm8	21786	21786	E1-CC2	–
qmof	20372	20372	outputs.pbe.bandgap	–
materials project	100000	99998	formation_energy_per_atom	–

Table 22: Retained entities, declared target slices and label coverage. Positive counts apply only to binary tasks; dashes denote continuous targets. Structural splitting retains valid features independently of missing responses. Single-target slices do not constitute an evaluation of all assays in a multitask source. Exact raw-file and retained-identity hashes accompany the catalogue.

13 Settings and computational scope

Setting	Power	Group threshold	Capacity tolerance
Plain	1	None	0, 0.01 or 0.05
Power only	3	None	0
Groups only	1	0.4	0
Combined	3	0.4	0, 0.01 or 0.05
Higher-threshold combined	3	0.6	0.01 or 0.05

Table 23: Readable setting names. Standard factors have rank 256, four restarts and at most 25 assignment iterations. Molecular thresholds refer to the 1024-bit radius-two Morgan/Tanimoto representation. Group caps are half the smallest target fold, with the freeze rule; top-32 neighbours use exact CPU blocks in the benchmark and exact GPU tiling in the H100 scaling study. Materials grouping instead uses a representation-specific similarity quantile.

Operation	Storage or arithmetic scope
Factor storage	$O(nr)$
Factor construction/whitening	$O(nrd + nr^2 + r^3)$ arithmetic
Fold products, fixed fold count	$O(nr)$ per iteration, plus assignment sorting
M point-refinement moves	Up to $O(Mnr)$ arithmetic
Exact neighbour construction	$O(n^2 d)$ arithmetic; blocked temporary storage
Retained neighbour indices	$O(nk)$ storage

Table 24: Computational scope for n entities, feature dimension d, rank r and neighbour degree k. Input features and outputs occupy additional storage. If the number of refinement moves grows with n, refinement can be quadratic. The factor-storage statement alone is not a linear-time claim.

14 Factor construction and optimization details

The main article presents the factor-space objective and its scientific use. This section records the numerical choices needed to reproduce the solver. The regularized landmark eigendecomposition floors eigenvalues at 10^{-6}.

Landmark selection affects approximation fidelity; fixed and adaptive sampling are established alternatives [Kumar et al., 2012, Musco and Musco, 2017]. The implementation supports uniform landmarks and a similarity-based spreading heuristic historically named `kmeans++`. The latter samples proportional to $(1 - \max_j S_{ij})^2$ over current landmarks. This is not ordinary squared Hilbert-distance sampling for a normalized kernel, so standard k-means++ initialization guarantees do not automatically apply. We retain its name for API compatibility and evaluate uniform sampling as an ablation.

The target count for each fold is obtained by largest-remainder rounding. For normalized requested fractions q_c, the tolerance corridor is based on $(1 \pm \epsilon)nq_c$, with integer bounds widened only enough to contain the rounded target. Zero tolerance yields the exact integer targets.

Each restart starts from a seeded count-constrained assignment. The two-fold linear subproblem is solved by sorting preference differences while enforcing both folds' bounds. Three-fold assignment uses a regret-ordered greedy peel that leaves sufficient capacity for the remaining folds. Proposals exclude self-similarity and are accepted only when the sum of squared fold sums does not decrease. A rejected proposal ends the restart. The best restart undergoes move refinement, or size-preserving pair swaps at exact capacities. The current fold is masked before alternative destinations are searched. Candidate limits make this a local search. Point-level refinement is skipped above two million entities. The standard settings are rank 256, four restarts and at most 25 assignment iterations.

For groups, the default freeze rule stops growth of the larger component after a cap-blocked merge. The alternative skip rule rejects only that merge. Weighted assignment and repair operate on group counts; groups dissolved into singletons are counted in the output diagnostics. Failure to satisfy the final count corridor raises an error. The multi-component extension concatenates per-axis factors. Dense identity columns for repeated featureless values can make its storage quadratic, so this extension is outside the reported scaling claims.

15 Neighbour construction details

The CPU reference scans exact similarities in row blocks. Blocking limits temporary storage while retaining quadratic arithmetic. Two-dimensional GPU tiling supplies a second exact backend. The optional NN-Descent candidate graph over packed binary fingerprints recomputes retained Tanimoto similarities using exact bit intersections. Its candidate recall is approximate. Top-32 recall and recall of exact top-32 edges at similarity at least 0.4 use 256 uniformly sampled query entities. Index-based recall is conservative when similarities tie. Compilation warm-up is recorded separately from warm graph construction. The retained graph-construction measurements appear below. Their split-quality and optimizer clocks are not used to characterize the revised solver.

16 Powered-kernel fidelity and rank sensitivity

The fidelity study uses six full datasets: FreeSolv, ESOL, BACE, BBBP, ClinTox and Lipophilicity. For each power, rank and seed, a factor and optimized partition are constructed with the release solver. Five count-preserving perturbations give a spectrum of partitions evaluated against the same factor and exact powered kernel. The tables distinguish empirical cut errors from the spectral residual estimate. The latter is obtained with a symmetric eigensolver, a fixed starting vector and tolerance 10^{-8}; its eigenpair residual is retained in the machine-readable results. It is not a certified numerical upper bound.

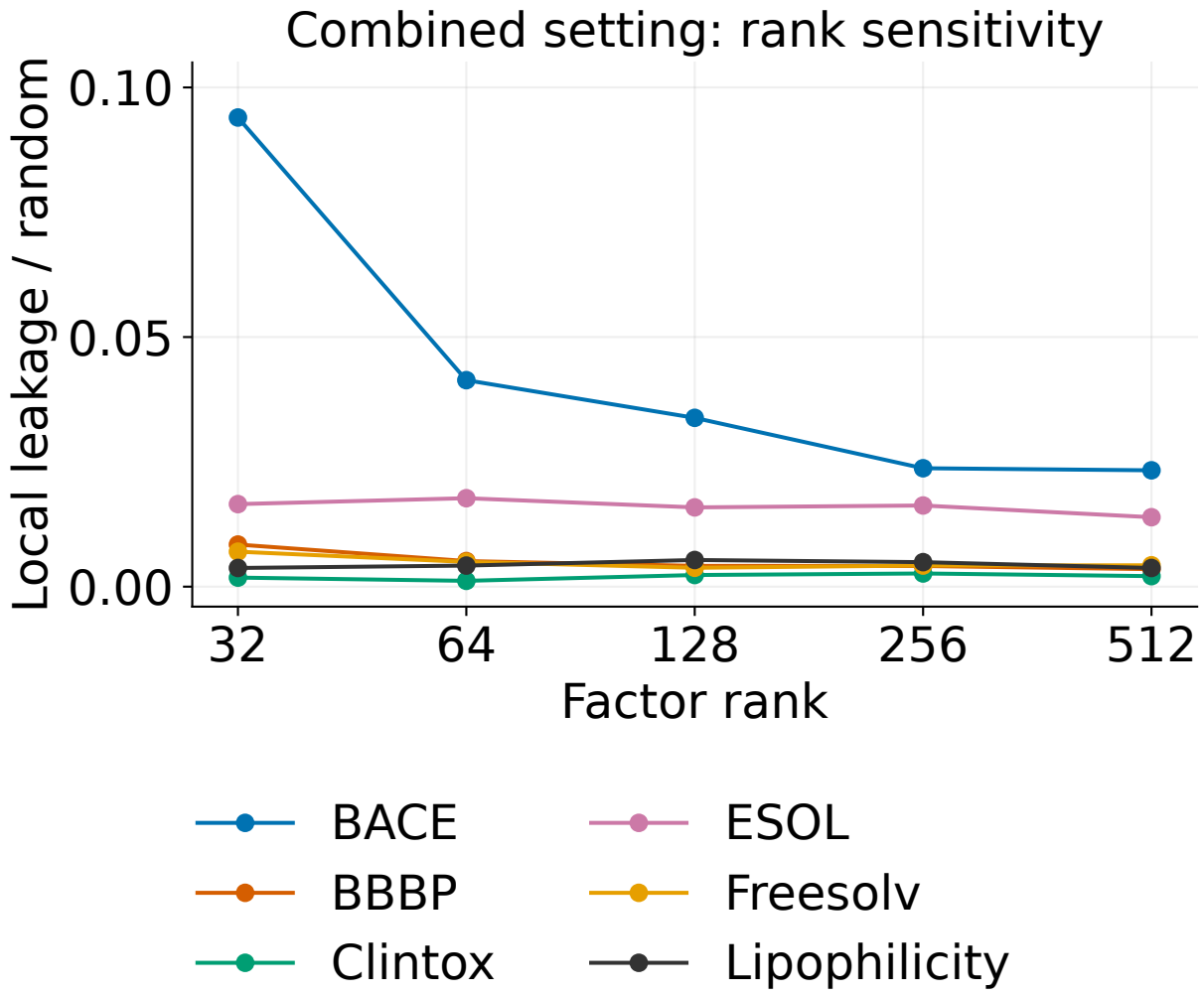


Figure 1: Combined-setting local leakage versus rank. Values average three split seeds at exact integer fold counts. Here local queries include validation and test, for continuity with the rank study's stored metric. The main baseline comparison uses test-only local queries and reports pooled held-out queries separately. Small nonmonotone changes are retained.

Entities	Backend	Graph seconds	Top-32 recall	Strong recall
1000	Exact tiled	0.009	1.000	1.000
10000	Exact tiled	0.240	1.000	1.000
100000	Exact tiled	13.002	1.000	1.000
30000	Exact tiled	1.287	1.000	1.000
1000	NN-Descent	0.842	1.000	1.000
10000	NN-Descent	8.132	0.993	1.000
100000	NN-Descent	97.569	0.990	0.998
30000	NN-Descent	26.129	0.994	1.000

Table 25: Neighbour-graph engineering measurements on ChEMBL fingerprints. Exact tiled search uses a Tesla T4; NN-Descent uses two CPU threads. Recall is measured on 256 sampled query entities; strong recall restricts exact top-32 edges to similarity at least 0.4. ANN compilation warm-up is separate. These are graph-construction observations from the preserved source study, not H100 split-call timings or measurements of the revised partition optimizer.

17 External assay sensitivity

Original and class-balanced partitions share fold counts and input molecules. Class-balanced assignments use target labels, and their structural composition changes. They diagnose sensitivity and are not a proposed label-blind splitter. Per-assay JSON records retain all assignments, input and producer hashes, nearest-training-molecule similarity and label support, and all model scores. Reported ROC-AUC below 0.5 is retained; no prediction is flipped after inspecting the test labels. Class rates accompany AUC so shifts are visible directly.

Power	Rank	Max error / mass	Median bound / mass	Max bound / mass
1	32	0.0808	0.4403	0.6789
1	64	0.0369	0.2488	0.3695
1	128	0.0164	0.1323	0.2451
1	256	0.0058	0.0716	0.1191
1	512	0.0021	0.0337	0.0569
3	32	0.3110	3.3021	7.2872
3	64	0.2788	3.1409	7.1525
3	128	0.2239	2.6336	7.0219
3	256	0.1563	2.1753	5.8603
3	512	0.0958	1.4566	2.7974

Table 26: Six full development datasets, three seeds and five partition perturbations per factor. Numerical spectral estimates use a symmetric eigensolver and are not certified upper bounds. Values above one give a vacuous bound on a cut ratio.

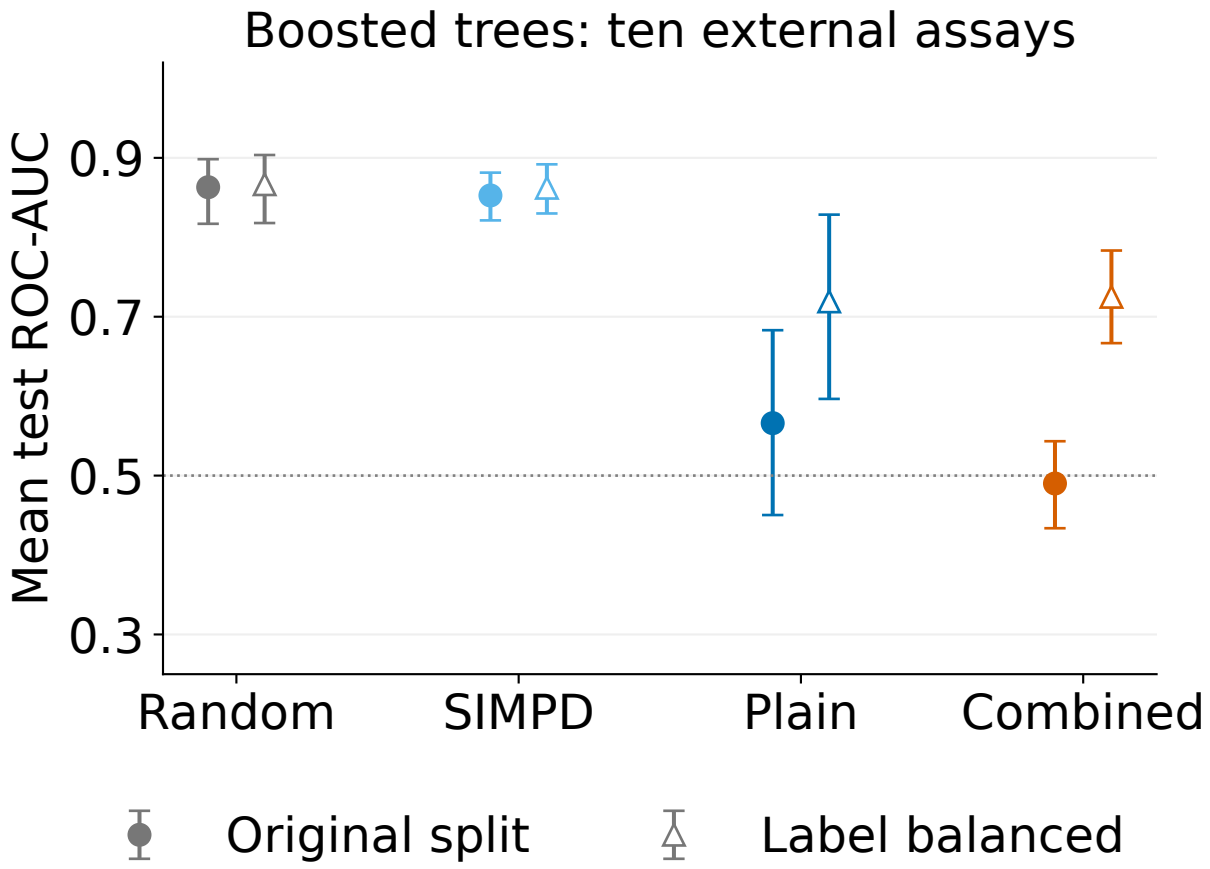


Figure 2: Boosted-tree absolute AUC with the same protocol and assay-bootstrap intervals as the main RF figure. Circles show original partitions; triangles show the target-aware intervention.

Assay	Random	SIMPD	Plain	Combined	Balanced
1293269-1	0.875	0.931	0.724	0.409	0.886
1889-1	0.870	0.903	0.496	0.464	0.846
225-1	0.784	0.848	0.534	0.525	0.583
237-1	0.890	0.839	0.605	0.529	0.656
269-1	0.933	0.887	0.500	0.384	0.867
2820-1	0.900	0.903	0.439	0.416	0.815
284-1	0.890	0.847	0.711	0.554	0.767
3837-1	0.932	0.901	0.456	0.410	0.545
3952-3	0.921	0.839	0.624	0.824	0.821
4105864-1	0.713	0.767	0.801	0.706	0.781

Table 27: RF mean test ROC-AUC by assay. Balanced refers to the combined partition after the target-aware intervention; it is not the label-blind method.

Assay	Random	SIMPD	Plain	Combined	Balanced
1293269-1	0.903	0.915	0.925	0.422	0.871
1889-1	0.858	0.912	0.701	0.601	0.841
225-1	0.795	0.849	0.476	0.596	0.663
237-1	0.883	0.846	0.647	0.543	0.586
269-1	0.927	0.867	0.512	0.499	0.776
2820-1	0.899	0.878	0.380	0.322	0.756
284-1	0.866	0.862	0.650	0.387	0.617
3837-1	0.909	0.861	0.515	0.560	0.620
3952-3	0.897	0.777	0.193	0.433	0.783
4105864-1	0.693	0.759	0.661	0.538	0.736

Table 28: XGB mean test ROC-AUC by assay. Balanced refers to the combined partition after the target-aware intervention; it is not the label-blind method.

Assay	Train positive rate	Test positive rate
1293269-1	0.415	0.356
1889-1	0.388	0.395
225-1	0.387	0.359
237-1	0.397	0.405
269-1	0.418	0.362
2820-1	0.403	0.396
284-1	0.412	0.390
3837-1	0.387	0.411
3952-3	0.394	0.352
4105864-1	0.389	0.451

Table 29: Class proportions for random; averages over split seeds.

Assay	Train positive rate	Test positive rate
1293269-1	0.417	0.350
1889-1	0.374	0.453
225-1	0.352	0.500
237-1	0.370	0.514
269-1	0.390	0.473
2820-1	0.391	0.444
284-1	0.388	0.486
3837-1	0.429	0.242
3952-3	0.363	0.479
4105864-1	0.410	0.367

Table 30: Class proportions for published simpd; averages over split seeds.

Assay	Train positive rate	Test positive rate
1293269-1	0.429	0.300
1889-1	0.374	0.453
225-1	0.383	0.375
237-1	0.433	0.257
269-1	0.468	0.161
2820-1	0.464	0.154
284-1	0.401	0.433
3837-1	0.386	0.414
3952-3	0.391	0.366
4105864-1	0.408	0.376

Table 31: Class proportions for plain; averages over split seeds.

Assay	Train positive rate	Test positive rate
1293269-1	0.333	0.683
1889-1	0.392	0.380
225-1	0.388	0.356
237-1	0.419	0.315
269-1	0.455	0.215
2820-1	0.466	0.145
284-1	0.400	0.438
3837-1	0.382	0.431
3952-3	0.430	0.211
4105864-1	0.430	0.287

Table 32: Class proportions for combined; averages over split seeds.

18 Scaling measurements

The plain-factor and exact-tiled combined pipelines use the same ChEMBL cache, nested sample indices, kernel rank and two-thread CPU allocation. Independent cases are distributed across reserved H100s, with one process per case. All returned assignments satisfy the stated two-sided count corridor. The table reports the actual measured ranges without extending the combined setting to unmeasured sizes. Saved sample indices and assignments allow independent checks of retention, counts and nesting.

Setting	Molecules	Seconds	GPU GiB	Host RSS GiB
Plain	1000	0.10	0.05	1.47
Plain	10000	0.32	0.16	2.12
Plain	100000	1.37	1.30	5.34
Plain	300000	3.55	3.83	11.86
Plain	1000000	10.83	12.67	34.70
Plain	2000000	19.93	25.32	67.34
Combined	1000	0.15	0.05	1.49
Combined	10000	1.67	0.16	2.13
Combined	30000	27.45	0.42	3.01
Combined	100000	89.04	1.30	5.33

Table 33: Fresh-process H100 NVL scaling cases at rank 256 and size tolerance 0.01. Split time excludes feature loading and scoring and follows a 100-entity warm-up. GPU peak is reset after warm-up; process RSS includes inputs and warm-up. The combined setting uses exact tiled neighbours. These are single-seed measurements, not repeated timing distributions.

19 Threshold and independent-representation robustness

Power/threshold cases and their controls share the same retained rows and exact 8:1:1 integer targets. Morgan and MACCS kernels are scored exactly on the full datasets, with test-only local queries and the complete training fold as reference. Three random assignments at those counts supply a common normalization for all settings within a seed. MACCS keys are never supplied to the optimizer. Its raw nearest-neighbour similarity is the primary independent representation check: the absolute Morgan ramp cutoffs have no universal interpretation for MACCS.

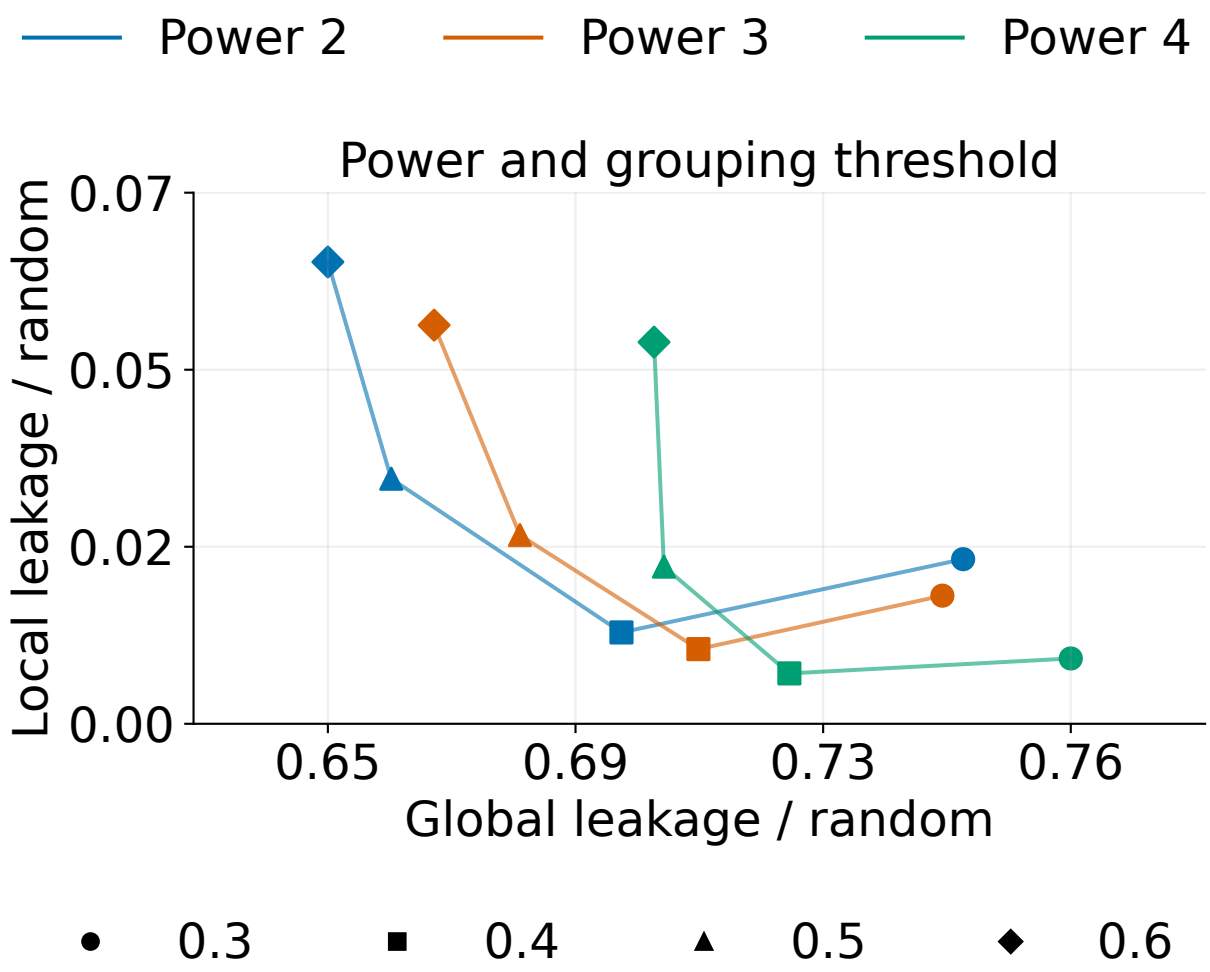


Figure 3: Power/threshold sensitivity at exact fold counts on six full development datasets. Each point averages dataset means after averaging three split seeds. Colours denote kernel power; marker shapes denote grouping thresholds, listed along the bottom. Lines connect thresholds in increasing order within each power. Both axes use the original Morgan kernel; local queries are test-only. The settings were not selected using external-assay prediction scores.

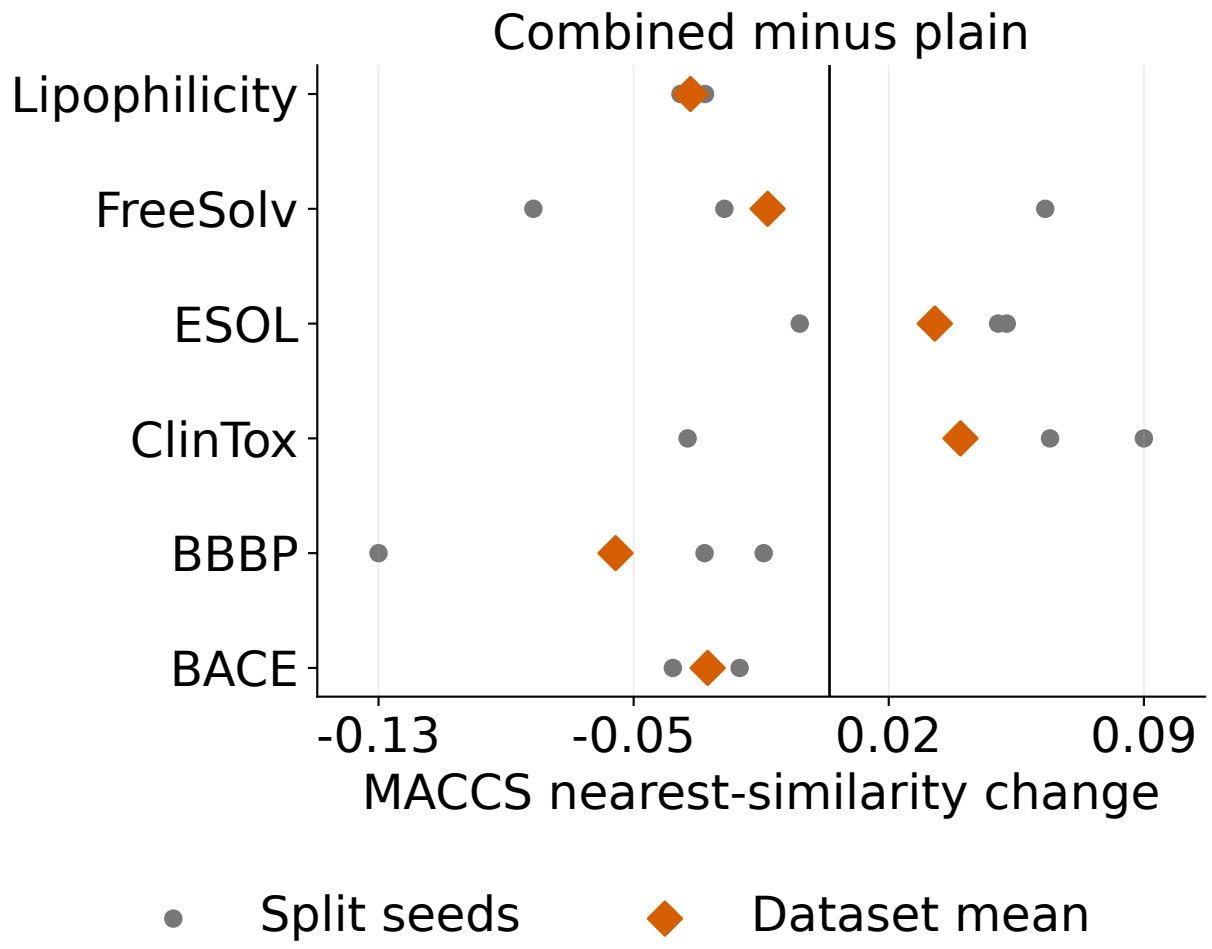


Figure 4: MACCS nearest-training-similarity changes for the fixed combined setting relative to plain low rank. Small circles are paired split-seed differences and diamonds are dataset means. Negative values indicate lower similarity. The two positive dataset means are retained.

Metric	Mean change	95% interval	Wins/ties	Sign p
Morgan ramp 0.4–0.8	-0.163	[-0.246, -0.086]	6/0	0.031
Morgan ramp 0.3–0.7	-0.201	[-0.308, -0.101]	6/0	0.031
Morgan ramp 0.5–0.9	-0.109	[-0.174, -0.054]	6/0	0.031
Morgan top-five ramp	-0.075	[-0.130, -0.020]	6/0	0.031
Morgan fraction above 0.6	-0.091	[-0.165, -0.024]	6/0	0.031
Morgan fraction above 0.8	-0.007	[-0.014, -0.001]	4/2	0.125
MACCS nearest similarity	-0.014	[-0.041, 0.015]	4/0	0.688
MACCS ramp 0.4–0.8	-0.002	[-0.060, 0.057]	4/0	0.688

Table 34: Combined minus plain on six full datasets after averaging three split seeds. Intervals are descriptive dataset bootstraps. Sign-test p-values are two-sided, exclude ties, and are unadjusted across correlated exploratory diagnostics. MACCS cutoffs are a sensitivity check, not a universal scale of chemical relatedness.

20 Component ablation

Six full datasets (BACE, BBBP, ESOL, FreeSolv, Lipophilicity and Tox21) are partitioned with the release solver at eighteen prespecified settings and three split seeds. This study separates landmark sampling, restart budget, move/swap refinement and cap policy in addition to the power, grouping, rank and capacity controls. It uses the same primary scoring protocol as the main benchmark: exact global scoring through 4,000 entities and a fixed uniform 4,000-row sample above that limit; test-only local queries against the full training fold, with three count-preserving permutation controls. All input entities are retained. The 0.05-tolerance row may change fold counts; the other rows use exact integer targets. Group dissolution counts describe output relaxations rather than a guarantee of hard analogue exclusion. NN-Descent can return fewer candidate neighbours for some samples; execution logs retain these native warnings, and the assignments are scored without tuning the search parameters after observing their results.

	Global ratio	Local ratio	Dissolved groups
Plain	0.750	0.315	–
Rank 32	0.756	0.497	–
Rank 64	0.752	0.398	–
Rank 128	0.751	0.408	–
Uniform landmarks	0.750	0.367	–
One restart	0.751	0.274	–
No move/swap refinement	0.750	0.323	–
Size tolerance 0.05	0.748	0.431	–
Power 2	0.761	0.123	–
Power 3	0.770	0.086	–
Groups only	0.777	0.149	1.000
Combined	0.786	0.022	1.000
Combined threshold 0.5	0.784	0.041	0.333
Combined threshold 0.6	0.777	0.037	0.000
Group cap 0.25	0.786	0.032	1.000
Group cap 1.0	0.789	0.013	1.000
Skip cap-blocked merges	0.791	0.240	1.333
Approximate neighbours	0.786	0.022	1.000

Table 35: bace: full release-code component ablation, averaged over three split seeds. Standard settings are rank 256, four restarts, move/swap refinement and exact counts. Combined uses power 3, threshold 0.4 and group cap 0.5. Each row changes the indicated setting; cap and neighbour rows modify Combined. Group dissolution is reported only for grouped settings. Actual fold counts and individual observations accompany the table.

	Global ratio	Local ratio	Dissolved groups
Plain	0.695	0.094	–
Rank 32	0.704	0.160	–
Rank 64	0.699	0.183	–
Rank 128	0.696	0.116	–
Uniform landmarks	0.695	0.177	–
One restart	0.696	0.126	–
No move/swap refinement	0.695	0.094	–
Size tolerance 0.05	0.685	0.112	–
Power 2	0.698	0.141	–
Power 3	0.709	0.082	–
Groups only	0.726	0.007	0.000
Combined	0.748	0.002	0.000
Combined threshold 0.5	0.715	0.021	0.000
Combined threshold 0.6	0.710	0.042	0.000
Group cap 0.25	0.734	0.000	0.000
Group cap 1.0	0.746	0.001	0.000
Skip cap-blocked merges	0.749	0.000	0.000
Approximate neighbours	0.748	0.002	0.000

Table 36: bbbp: full release-code component ablation, averaged over three split seeds. Standard settings are rank 256, four restarts, move/swap refinement and exact counts. Combined uses power 3, threshold 0.4 and group cap 0.5. Each row changes the indicated setting; cap and neighbour rows modify Combined. Group dissolution is reported only for grouped settings. Actual fold counts and individual observations accompany the table.

	Global ratio	Local ratio	Dissolved groups
Plain	0.528	0.099	–
Rank 32	0.532	0.146	–
Rank 64	0.530	0.154	–
Rank 128	0.529	0.151	–
Uniform landmarks	0.528	0.117	–
One restart	0.528	0.109	–
No move/swap refinement	0.528	0.087	–
Size tolerance 0.05	0.519	0.089	–
Power 2	0.548	0.128	–
Power 3	0.608	0.069	–
Groups only	0.553	0.023	0.000
Combined	0.644	0.023	0.000
Combined threshold 0.5	0.604	0.041	0.000
Combined threshold 0.6	0.579	0.075	0.000
Group cap 0.25	0.643	0.022	0.000
Group cap 1.0	0.631	0.013	0.667
Skip cap-blocked merges	0.646	0.021	0.000
Approximate neighbours	0.644	0.023	0.000

Table 37: esol: full release-code component ablation, averaged over three split seeds. Standard settings are rank 256, four restarts, move/swap refinement and exact counts. Combined uses power 3, threshold 0.4 and group cap 0.5. Each row changes the indicated setting; cap and neighbour rows modify Combined. Group dissolution is reported only for grouped settings. Actual fold counts and individual observations accompany the table.

	Global ratio	Local ratio	Dissolved groups
Plain	0.420	0.122	–
Rank 32	0.429	0.193	–
Rank 64	0.423	0.284	–
Rank 128	0.420	0.140	–
Uniform landmarks	0.420	0.228	–
One restart	0.420	0.154	–
No move/swap refinement	0.420	0.123	–
Size tolerance 0.05	0.406	0.011	–
Power 2	0.431	0.079	–
Power 3	0.519	0.056	–
Groups only	0.501	0.004	0.000
Combined	0.505	0.006	0.333
Combined threshold 0.5	0.485	0.010	0.000
Combined threshold 0.6	0.455	0.051	0.000
Group cap 0.25	0.502	0.006	0.000
Group cap 1.0	0.506	0.010	0.000
Skip cap-blocked merges	0.510	0.002	0.000
Approximate neighbours	0.505	0.009	0.000

Table 38: freesolv: full release-code component ablation, averaged over three split seeds. Standard settings are rank 256, four restarts, move/swap refinement and exact counts. Combined uses power 3, threshold 0.4 and group cap 0.5. Each row changes the indicated setting; cap and neighbour rows modify Combined. Group dissolution is reported only for grouped settings. Actual fold counts and individual observations accompany the table.

	Global ratio	Local ratio	Dissolved groups
Plain	0.801	0.335	–
Rank 32	0.811	0.359	–
Rank 64	0.805	0.423	–
Rank 128	0.803	0.320	–
Uniform landmarks	0.801	0.298	–
One restart	0.801	0.317	–
No move/swap refinement	0.802	0.319	–
Size tolerance 0.05	0.796	0.307	–
Power 2	0.804	0.300	–
Power 3	0.809	0.272	–
Groups only	0.865	0.004	0.000
Combined	0.871	0.004	0.000
Combined threshold 0.5	0.820	0.019	0.000
Combined threshold 0.6	0.815	0.060	0.000
Group cap 0.25	0.845	0.003	0.000
Group cap 1.0	0.872	0.003	0.000
Skip cap-blocked merges	0.875	0.002	0.000
Approximate neighbours	0.871	0.004	0.000

Table 39: lipophilicity: full release-code component ablation, averaged over three split seeds. Standard settings are rank 256, four restarts, move/swap refinement and exact counts. Combined uses power 3, threshold 0.4 and group cap 0.5. Each row changes the indicated setting; cap and neighbour rows modify Combined. Group dissolution is reported only for grouped settings. Actual fold counts and individual observations accompany the table.

	Global ratio	Local ratio	Dissolved groups
Plain	0.573	0.210	–
Rank 32	0.579	0.299	–
Rank 64	0.577	0.182	–
Rank 128	0.572	0.152	–
Uniform landmarks	0.574	0.172	–
One restart	0.573	0.151	–
No move/swap refinement	0.573	0.217	–
Size tolerance 0.05	0.565	0.175	–
Power 2	0.588	0.264	–
Power 3	0.607	0.250	–
Groups only	0.655	0.105	0.000
Combined	0.721	0.025	0.000
Combined threshold 0.5	0.651	0.045	0.000
Combined threshold 0.6	0.623	0.116	0.000
Group cap 0.25	0.719	0.023	0.000
Group cap 1.0	0.698	0.023	0.000
Skip cap-blocked merges	0.715	0.023	0.000
Approximate neighbours	0.721	0.025	0.000

Table 40: tox21: full release-code component ablation, averaged over three split seeds. Standard settings are rank 256, four restarts, move/swap refinement and exact counts. Combined uses power 3, threshold 0.4 and group cap 0.5. Each row changes the indicated setting; cap and neighbour rows modify Combined. Group dissolution is reported only for grouped settings. Actual fold counts and individual observations accompany the table.

21 Published local and cluster baselines

Lo-Hi uses the official Hi objective, threshold 0.4 and no coarsening. Resource controls cap it at two CPU threads and 7,200 solver seconds, with the native solver outcome retained. Eligible full datasets have at most 5,000 molecules. The comparison preserves the native retained subset and exact realized counts; discarded molecules are never assigned to a test fold by another name.

The UMAP recipe follows the published NCI-60 clustering study [Hernández-Hernández and Ballester, 2023]: radius-two 1024-bit Morgan fingerprints, Jaccard UMAP with 100 neighbours, minimum distance zero, two components and seed 42, followed by Ward agglomeration into seven clusters. We transfer this recipe to new datasets and evaluate all seven native cluster holdouts. Low-rank controls exactly match each holdout's counts and retain all rows. The memory policy excludes cases whose condensed float64 agglomeration distances alone exceed 16 GiB. FreeSolv instead exhibits a native method failure: one valid fingerprint is disconnected in Jaccard similarity, producing a nonfinite embedding coordinate rejected by Ward clustering. We neither impute nor drop the row to manufacture a result.

Dataset	Plain G	Plain N	Combined G	Combined N	Lo-Hi G	Lo-Hi N
bace	0.729	0.314	0.764	0.004	0.948	0.000
bbbp	0.607	0.126	0.640	0.000	0.988	0.000
clintox	0.562	0.162	0.595	0.000	0.932	0.000
esol	0.432	0.107	0.533	0.012	0.933	0.000
freesolv	0.309	0.128	0.412	0.000	0.801	0.000
lipophilicity	0.756	0.318	0.805	0.003	0.974	0.000
sider	0.466	0.117	0.494	0.000	0.989	0.000

Table 41: Lo-Hi comparisons use each native retained subset and its exact train/test counts. Low-rank settings average three seeds. Retained fractions and native solver states are reported separately. G and N denote global and local ratios, evaluated within each study protocol.

Dataset	Plain G	Plain N	Combined G	Combined N	UMAP G	UMAP N
bace	0.678	0.224	0.726	0.047	0.863	0.034
bbbp	0.605	0.124	0.638	0.000	0.923	0.012
clintox	0.585	0.142	0.635	0.001	0.922	0.042
esol	0.448	0.108	0.546	0.016	0.843	0.063
hiv	0.581	0.223	0.715	0.023	0.983	0.109
lipophilicity	0.757	0.295	0.810	0.003	0.951	0.027
qm8	0.573	0.315	0.684	0.043	0.893	0.053
sider	0.466	0.125	0.495	0.000	0.913	0.031
tox21	0.503	0.228	0.621	0.025	0.888	0.143

Table 42: UMAP comparisons average all seven native cluster holdouts equally within each dataset. Low-rank controls retain all entities and exactly match each holdout count. FreeSolv is an audited native embedding/clustering failure; MUV exceeds the prespecified memory policy. Neither is assigned a score. G and N denote global and local ratios, evaluated within each study protocol.

22 Graph-degree and optimization sensitivity

Graph degree is varied over 16, 32, 64 and 128 on six full development datasets with three seeds and tolerance 0.01. The prespecified primary comparator uses 32 neighbours, matching the grouping builder's degree. All settings use the ordinary pairwise Graph API, quality preset and CUT objective. Larger graphs retain more dense similarity mass but change the global–local trade-off; no single degree is declared best for both metrics. Native and repaired assignments are preserved. Tables distinguish degree sensitivity from the additional effect of imposing both lower and upper capacity bounds.

The FreeSolv search control compares four and thirty restarts at exact counts for plain and combined settings, with five fixed seeds. A small change under this finite search budget is empirical saturation, not a lower bound on the unknown optimum. The combined setting shows larger objective changes than the plain setting, so the earlier development claim that extra restarts have no effect is not carried forward.

Neighbours	Retained mass	Global ratio	Local ratio
16	0.0624	0.8218	0.0138
32	0.1057	0.8103	0.0184
64	0.1749	0.7639	0.0538
128	0.2824	0.7166	0.0296

Table 43: Graph degree sensitivity on six full datasets. Dataset means average three seeds before cross-dataset averaging. Retained mass uses quantized graph weights and the full ordinary-kernel mass, with quantization-error bounds supplied in the CSV. Global and local scores follow the main common protocol. Build, partition and capacity repair are included in split time.

Setting	Dense cut change	Powered cut change	Local ratio change
combined	-0.0182	-0.0297	0.0214
plain	-0.0004	-0.0004	-0.0687

Table 44: Thirty versus four restarts on full FreeSolv at exact counts, averaged over five fixed seeds. Cut changes are relative to the four-restart value; local changes are absolute. Both use the same factor seed within each pair. More restarts do not certify optimality, and choosing a different pre-refinement basin can change the final polished solution.

23 Materials generality check

Leaving related materials together is a recognized evaluation choice [Meredig et al., 2018, Witman and Schindler, 2025], and structure-based out-of-distribution benchmarks test transfer beyond random holdouts [Omee et al., 2024]. Here the QMOF and Materials Project checks use composition descriptors and a standardized Euclidean similarity kernel. They test applicability of the factor formulation beyond molecular Tanimoto similarity, rather than crystal-structure novelty. Full per-dataset scores accompany the benchmark tables. The study makes no empirical claim about untested sequence or biomolecular featurizers.

Dataset	Plain G	Plain N	Combined G	Combined N	DataSAIL G	DataSAIL N
bace	0.750	0.363	0.785	0.066	0.775	0.007
bbbp	0.692	0.099	0.743	0.001	0.731	0.002
clintox	0.658	0.082	0.698	0.001	0.699	0.020
esol	0.524	0.093	0.635	0.027	0.597	0.072
freesolv	0.415	0.357	0.497	0.009	0.510	0.000
lipophilicity	0.799	0.314	0.869	0.004	0.956	0.013
sider	0.594	0.091	0.625	0.000	0.634	0.000

Table 45: Exact-count controls for saved DataSAIL assignments at nominal tolerance 0.01. Each low-rank partition retains the same entities and has exactly the three native DataSAIL counts. Low-rank scores average three seeds; DataSAIL has one assignment per configuration. G and N denote global and test-only local ratios. These reused assignments supply no new runtime.

Recorded outcome	Tolerance 0.01	Tolerance 0.05
feasible assignment; no optimality certificate	3	3
no feasible assignment returned within configured solve	1	1
no recorded attempt under this protocol	3	3
resource-policy exclusion	2	2
solver-reported optimum of clustered problem	4	4

Table 46: DataSAIL source coverage, classified from native logs and saved assignments. Solver-reported optima concern its clustered integer program. The legacy CSV label "infeasible" is not treated as proof of infeasibility when no such solver certificate was retained. Resource exclusions and missing recorded attempts are separate.

24 Calibration protocol and prediction provenance

The hardness parameter uses a nested seeded sequence of point swaps between an optimized assignment and its count-preserving random endpoint. Its observed predictive response need not be monotone. Classification difficulty is $\mathrm{AUC}_0 - \mathrm{AUC}_\alpha$; regression difficulty is $\mathrm{RMSE}_\alpha/\mathrm{RMSE}_0 - 1$. Subscript zero denotes the random endpoint with the same split seed,

fold counts and model. These definitions use test scores and avoid interpreting a nearly perfect RF training score as a separate measure of difficulty.

Five knots, $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$, are fitted with RF on split seeds 0–2. The mean curve uses isotonic regression followed by piecewise-linear interpolation. Its direction is estimated from the calibration probes. Fitting can pool noisy negative early differences, so the fitted random endpoint can differ slightly from the known observed value zero. Seeds 10–14 are excluded from that fit and evaluate all five knots, as well as three interior targets at 25%, 50% and 75% of the fitted difficulty range. RF and boosted trees each have model initialization seed zero. The study has 570 model fits and 378 saved assignments across six datasets. Observed adjacent-knot decreases are reported before isotonic fitting; a fitted monotone curve is not evidence that individual predictive responses are monotone.

Fixed-knot validation and interior-target inversion evaluate different points. Their errors therefore receive separate tables and source columns. The random-relative study's authoritative summary is `calibration_v2/validation_summary.csv`. The older `calibration_validation.csv` and `inversion_validation.csv` describe 25 fixed-knot and 15 interior-target scores per dataset/model, respectively, using train-minus-test gaps. Recomputed legacy errors match both files; they are separate studies and are not used as random-relative calibration errors. The source reconciliation is included with the release records.

Dataset	Units	RF MAE	XGB MAE
bace	AUC decrement	0.041	0.023
bbbp	AUC decrement	0.052	0.047
esol	relative RMSE	0.092	0.096
freesolv	relative RMSE	0.555	0.624
lipophilicity	relative RMSE	0.044	0.083
sider	AUC decrement	0.030	0.042

Table 47: Fixed-knot validation on five unseen split seeds (25 points per model and dataset). Classification uses AUC(random) minus AUC(split); regression uses RMSE(split)/RMSE(random) minus one. No test label is used to fit the calibration on validation seeds. Each model has initialization seed zero.

Dataset	Units	RF MAE	XGB MAE
bace	AUC decrement	0.056	0.030
bbbp	AUC decrement	0.062	0.054
esol	relative RMSE	0.101	0.115
freesolv	relative RMSE	0.711	0.747
lipophilicity	relative RMSE	0.060	0.096
sider	AUC decrement	0.067	0.069

Table 48: Interior-target inversion on the same five validation seeds (15 points per model and dataset). Targets are fitted using RF calibration seeds 0–2. Classification uses AUC(random) minus AUC(split); regression uses RMSE(split)/RMSE(random) minus one. No test label is used to fit the calibration on validation seeds. Each model has initialization seed zero.

Dataset	Model	Decreases	Adjacent steps
bace	rf	6	20
bace	xgb	3	20
bbbp	rf	6	20
bbbp	xgb	5	20
esol	rf	8	20
esol	xgb	4	20
freesolv	rf	4	20
freesolv	xgb	5	20
lipophilicity	rf	0	20
lipophilicity	xgb	1	20
sider	rf	4	20
sider	xgb	8	20

Table 49: Observed decreases in random-relative difficulty as hardness increases on validation seeds. Four adjacent knot steps are checked for each of five seeds. Isotonic fitting does not remove these violations from the observed scores.

The D-MPNN metric-validation study reuses saved predictions from the specified 30-epoch protocol and five model initializations. Test SMILES are checked against the prediction rows before ensembling. Its split assignments and prior group-cap rule remain attached to those measurements. It is a supporting metric study, separate from the release-code thirteen-dataset benchmark. Per-model and per-split variability is reported at its actual level of replication.

25 Metric-validation provenance

The synthetic analysis excludes random-reference and all hypergraph rows. Global and local correlations use the same finite observations. Correlations by constructed regime and all included rows are distributed as CSV files; pooled associations can also reflect differences between regimes.

Per-molecule validation uses existing five-initialization D-MPNN predictions for the random and plain-low-rank partitions of four datasets. Prediction rows are checked against test SMILES before ensembling. Training-only references are primary. Training-plus-validation references are retained as a secondary check. Within-split correlations accompany pooled results because the latter also mix split membership with molecular similarity. These analyses validate metric interpretation; they are not new release-code neural-model experiments.

Regime	Observations	Global rho	Local rho
continuum	24	0.844	0.578
hierarchy	30	0.402	0.491
long_tail	30	-0.022	0.631
near_dup	24	-0.290	0.588
pooled	120	0.020	0.519
tightness	12	0.435	0.808

Table 50: Family-leakage correlations on identical finite observations for both metrics. Local similarity is more strongly associated in four of the five constructed regimes; the continuum regime favors the global measure. The pooled contrast also reflects differences among regimes.

Dataset	Partition	Test entities	NN rho	Global rho
BBBP	Plain	387	0.064	0.209
BBBP	Random	408	-0.078	-0.142
ESOL	Plain	214	-0.189	-0.074
ESOL	Random	226	-0.094	0.078
FreeSolv	Plain	121	-0.375	-0.072
FreeSolv	Random	128	-0.109	-0.042
Lipophilicity	Plain	798	-0.187	-0.058
Lipophilicity	Random	840	-0.217	0.039

Table 51: Within-partition correlations with absolute five-initialization D-MPNN ensemble error. NN similarity uses the actual training fold as reference; the global proxy is the per-point overlap score. Each partition has one split realization. Pooled and training-plus-validation analyses accompany these primary within-partition values as CSV files.

26 Complete benchmark tables

Nominal size tolerance 0.01

	Global ratio	Local ratio	Split seconds
Random	1.003	0.914	0.009
Plain	0.420	0.123	0.132
Combined	0.504	0.009	0.187
Higher threshold	0.455	0.051	0.235
k-NN graph	0.547	0.026	0.324
Scaffold	0.760	0.240	0.052
Butina	0.826	0.261	0.097
DataSAIL	0.510	0.000	–

Table 52: freesolv at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.005	1.013	0.013
Plain	0.526	0.087	0.169
Combined	0.641	0.021	0.305
Higher threshold	0.581	0.068	0.379
k-NN graph	0.727	0.018	0.410
Scaffold	0.929	0.469	0.156
Butina	0.881	0.215	0.319
DataSAIL	0.597	0.072	–

Table 53: esol at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

Nominal size tolerance 0.05

	Global ratio	Local ratio	Split seconds
Random	0.994	0.977	0.012
Plain	0.750	0.373	0.207
Combined	0.784	0.016	0.360
Higher threshold	0.776	0.035	0.317
k-NN graph	0.918	0.035	0.410
Scaffold	0.981	0.865	0.621
Butina	0.861	0.346	0.764
DataSAIL	0.775	0.007	–

Table 54: bace at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.000	1.011	0.015
Plain	0.693	0.152	0.266
Combined	0.744	0.003	0.591
Higher threshold	0.707	0.038	0.558
k-NN graph	0.887	0.011	0.533
Scaffold	1.013	0.517	0.577
Butina	0.924	0.142	1.103
DataSAIL	0.731	0.002	–

Table 55: bbbp at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.001	0.996	0.011
Plain	0.596	0.084	0.206
Combined	0.623	0.000	0.425
Higher threshold	0.615	0.040	0.453
k-NN graph	0.793	0.007	0.463
Scaffold	1.037	0.564	2.103
Butina	0.934	0.126	0.643
DataSAIL	0.634	0.000	–

Table 56: sider at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.999	0.959	0.010
Plain	0.660	0.071	0.199
Combined	0.698	0.001	0.435
Higher threshold	0.682	0.063	0.439
k-NN graph	0.834	0.003	0.476
Scaffold	1.022	0.527	0.507
Butina	0.931	0.117	0.626
DataSAIL	0.699	0.020	–

Table 57: clintox at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.001	1.006	0.033
Plain	0.802	0.318	0.447
Combined	0.869	0.004	1.788
Higher threshold	0.814	0.062	1.284
k-NN graph	0.948	0.017	1.027
Scaffold	0.983	0.651	1.380
Butina	0.945	0.111	4.740
DataSAIL	0.956	0.013	–

Table 58: lipophilicity at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.016	1.003	0.075
Plain	0.571	0.218	0.774
Combined	0.722	0.025	4.120
Higher threshold	0.623	0.117	3.507
k-NN graph	0.750	0.018	2.462
Scaffold	0.959	0.454	1.582
Butina	0.920	0.235	16.415
DataSAIL	–	–	–

Table 59: tox21 at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.002	1.008	0.329
Plain	0.640	0.595	3.673
Combined	0.703	0.012	17.568
Higher threshold	0.670	0.120	40.124
k-NN graph	0.686	0.006	13.593
Scaffold	0.941	0.603	2.029
Butina	–	–	–
DataSAIL	–	–	–

Table 60: qm8 at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.953	0.996	223.143
Plain	0.688	0.205	13.679
Combined	0.832	0.029	70.735
Higher threshold	0.706	0.124	98.239
k-NN graph	0.879	0.051	41.908
Scaffold	0.987	0.573	12.942
Butina	–	–	–
DataSAIL	–	–	–

Table 61: hiv at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.979	0.999	1.330
Plain	0.779	0.516	14.229
Combined	0.907	0.256	261.990
Higher threshold	0.840	0.288	974.538
k-NN graph	0.949	0.109	364.558
Scaffold	0.972	0.748	32.771
Butina	–	–	–
DataSAIL	–	–	–

Table 62: muv at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.991	1.003	0.075
Plain	0.735	0.179	1.572
Combined	0.761	0.028	10.763
Higher threshold	0.761	0.028	11.139
k-NN graph	0.818	0.005	5.665
Scaffold	–	–	–
Butina	–	–	–
DataSAIL	–	–	–

Table 63: qmof at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.991	0.997	0.660
Plain	0.783	0.299	7.136
Combined	0.819	0.144	129.969
Higher threshold	0.819	0.144	125.466
k-NN graph	0.869	0.192	71.073
Scaffold	–	–	–
Butina	–	–	–
DataSAIL	–	–	–

Table 64: materials project at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.003	0.914	0.005
Plain	0.406	0.011	0.134
Combined	0.490	0.009	0.188
Higher threshold	0.440	0.040	0.248
k-NN graph	0.547	0.027	0.335
Scaffold	0.760	0.240	0.052
Butina	0.826	0.261	0.097
DataSAIL	0.494	0.023	–

Table 65: freesolv at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.005	1.013	0.012
Plain	0.519	0.089	0.170
Combined	0.626	0.024	0.299
Higher threshold	0.572	0.077	0.362
k-NN graph	0.668	0.022	0.381
Scaffold	0.929	0.469	0.160
Butina	0.881	0.215	0.319
DataSAIL	0.588	0.073	–

Table 66: esol at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.994	0.977	0.011
Plain	0.748	0.431	0.215
Combined	0.776	0.008	0.343
Higher threshold	0.771	0.021	0.321
k-NN graph	0.915	0.042	0.417
Scaffold	0.981	0.865	0.617
Butina	0.861	0.346	0.754
DataSAIL	0.763	0.012	–

Table 67: bace at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.000	1.011	0.015
Plain	0.685	0.112	0.248
Combined	0.736	0.001	0.593
Higher threshold	0.700	0.038	0.561
k-NN graph	0.887	0.010	0.575
Scaffold	1.013	0.517	0.571
Butina	0.924	0.142	1.075
DataSAIL	0.722	0.002	–

Table 68: bbbp at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.001	0.996	0.011
Plain	0.584	0.110	0.205
Combined	0.611	0.000	0.370
Higher threshold	0.605	0.050	0.419
k-NN graph	0.797	0.008	0.493
Scaffold	1.037	0.564	2.096
Butina	0.934	0.126	0.656
DataSAIL	0.620	0.000	–

Table 69: sider at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.999	0.959	0.011
Plain	0.650	0.074	0.200
Combined	0.688	0.001	0.402
Higher threshold	0.673	0.057	0.467
k-NN graph	0.837	0.021	0.499
Scaffold	1.022	0.527	0.491
Butina	0.931	0.117	0.633
DataSAIL	0.690	0.021	–

Table 70: clintox at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.001	1.006	0.034
Plain	0.796	0.307	0.458
Combined	0.868	0.004	1.711
Higher threshold	0.810	0.072	1.269
k-NN graph	0.918	0.025	1.032
Scaffold	0.983	0.651	1.370
Butina	0.945	0.111	4.701
DataSAIL	0.985	0.010	–

Table 71: lipophilicity at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.016	1.003	0.078
Plain	0.565	0.175	0.873
Combined	0.710	0.025	3.902
Higher threshold	0.616	0.116	3.685
k-NN graph	0.747	0.024	2.376
Scaffold	0.959	0.454	1.590
Butina	0.920	0.235	16.277
DataSAIL	–	–	–

Table 72: tox21 at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.002	1.008	0.321
Plain	0.636	0.618	2.929
Combined	0.799	0.083	544.601
Higher threshold	0.669	0.130	31.234
k-NN graph	0.681	0.009	13.487
Scaffold	0.941	0.603	2.013
Butina	–	–	–
DataSAIL	–	–	–

Table 73: qm8 at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.953	0.996	5.400
Plain	0.685	0.243	6.900
Combined	0.830	0.033	75.497
Higher threshold	0.701	0.139	102.224
k-NN graph	0.853	0.048	44.556
Scaffold	0.987	0.573	12.903
Butina	–	–	–
DataSAIL	–	–	–

Table 74: hiv at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.979	0.999	1.286
Plain	0.774	0.515	21.295
Combined	0.915	0.238	243.311
Higher threshold	0.833	0.290	1075.031
k-NN graph	0.949	0.111	359.014
Scaffold	0.972	0.748	40.553
Butina	–	–	–
DataSAIL	–	–	–

Table 75: muv at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.991	1.003	0.055
Plain	0.729	0.182	1.425
Combined	0.752	0.028	12.158
Higher threshold	0.752	0.028	12.082
k-NN graph	0.786	0.007	6.095
Scaffold	–	–	–
Butina	–	–	–
DataSAIL	–	–	–

Table 76: qmof at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.991	0.997	0.246
Plain	0.781	0.281	7.325
Combined	0.810	0.130	121.645
Higher threshold	0.810	0.130	122.091
k-NN graph	0.915	0.187	75.454
Scaffold	–	–	–
Butina	–	–	–
DataSAIL	–	–	–

Table 77: materials project at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

Comparison	Metric	n	Mean change	95% interval	Holm p
Plain / k-NN graph	Global	11	-0.163	[-0.18, -0.13]	0.020
Plain / k-NN graph	Local	11	0.222	[0.13, 0.32]	0.020
Combined / Plain	Global	11	0.082	[0.06, 0.11]	0.020
Combined / Plain	Local	11	-0.215	[-0.31, -0.14]	0.020
Combined / k-NN graph	Global	11	-0.081	[-0.11, -0.05]	0.152
Combined / k-NN graph	Local	11	0.007	[-0.01, 0.04]	1.000
Plain / DataSAIL	Global	7	-0.067	[-0.10, -0.04]	0.156
Plain / DataSAIL	Local	7	0.184	[0.09, 0.28]	0.156
Combined / DataSAIL	Global	7	-0.007	[-0.04, 0.02]	1.000
Combined / DataSAIL	Local	7	-0.001	[-0.02, 0.02]	1.000

Table 78: Paired molecular-dataset effects at tolerance 0.01. Changes subtract the second method from the first. Dataset-bootstrap intervals use 10,000 resamples after averaging split seeds. Sign-test p values receive Holm correction across the 20 declared comparisons (five method pairs, two metrics and two tolerances). DataSAIL pairs use exact-count controls. These convenience datasets support descriptive, representation-specific inference.

27 Same-host timing details

Dataset	Method	Median seconds	Global ratio	Local ratio
bace	combined	0.555	0.785	0.066
bace	datasail	7210.760	0.766	0.019
bace	plain	0.279	0.750	0.363
bbbp	combined	1.041	0.743	0.001
bbbp	datasail	7219.393	0.731	0.002
bbbp	plain	0.337	0.692	0.099
clintox	combined	1.057	0.698	0.001
clintox	datasail	1411.427	0.699	0.020
clintox	plain	0.434	0.658	0.082
lipophilicity	combined	2.700	0.869	0.004
lipophilicity	datasail	8796.726	0.956	0.013
lipophilicity	plain	1.095	0.799	0.314

Table 79: Same-host split-call time and exact-count quality. DataSAIL distance construction, clustering and integer programming are inside its clock; imports, feature loading and scoring are outside. Low-rank calls exclude a 100-entity warm-up. DataSAIL has one solve per dataset; low-rank entries aggregate three seeds. The solver time limit is 7,200 seconds and is not an end-to-end wall-time bound. Native solver states and all three low-rank observations accompany the table.

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